

# Independent external validation of the SugarCode AI computational-biology platform against held-out public benchmarks

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Independent research; computational work assisted by Instinct

## Abstract

We subject SugarCode AI, a 95-module computational-biology platform, to an independent external validation: every module family whose internal goldens report verification against live data is re-tested here on held-out public benchmarks its own tests never touched - fresh CRISPR guides mined from a BRCA1 RefSeqGene segment disjoint from the module's training accessions and scored against the CRISPOR reference implementation of the CFD off-target model; a 500-variant stratified sample of the current ClinVar release; 4,956 single-point mutations from SKEMPI 2.0; the BiGG *E. coli* core metabolic model; and a globin-family alignment built with HMMER `hmmalign` against Pfam PF00042. We report per-module agreement metrics with bootstrap confidence intervals, publish all fixtures with provenance, and document failure modes without suppressing negative results.

## 1 Introduction

Computational-biology toolkits are typically validated against the same data that informed their development, inflating confidence. Independent, held-out validation against public reference implementations and current database releases is the standard this platform must meet to be usable in laboratory settings. This study asks a single question: does each verified claim in the platform's own status ledger survive contact with data it has never seen? We answer it module family by module family, with frozen fixtures, pinned random seeds, and metrics reported with 95% bootstrap confidence intervals. All code, fixtures, and results are released in this repository.

## 2 Related Work and Benchmarks

Reference implementations used here are CRISPOR for guide scoring [2] and its Doench Rule Set 1/CFD models [1], SKEMPI 2.0 for binding free-energy changes [3], BiGG for metabolic models [4], Clustal Omega for alignment [5], and the Tavtigian Bayesian ACMG framework [6]. Pharmacogenomic tables come from the CPIC API; RNA structure references from Rfam seed alignments; splice predictions are compared with SpliceAI.

### 3 Methods

For each module family we constructed a held-out test disjoint from every fixture used during platform development. Negative results are preserved and reported alongside positives; no test was re-designed after seeing its outcome.

#### 3.1 Data and Provenance

All datasets are public and carry recorded provenance (see `data/README.md`): NCBI ClinVar `variant_summary.txt.gz` (retrieved 2026-09-24; 500-variant stratified reservoir sample from 1,517,551 eligible GRCh38 SNVs with criteria-provided review status); SKEMPI 2.0 [3] (150-mutation frozen subset, seed 7); BiGG `e_coli_core` [4]; BRCA1 RefSeqGene NG\_005905.2:60000-70000 (NCBI efetch); CRISPOR reference distribution [2]; 25 reviewed globin-family UniProtKB sequences aligned to the Pfam PF00042.29 HMM with HMMER3 `hmmalign` (pyhmmmer 0.12.3); the EBI Clustal Omega job queued for over 40 minutes and was abandoned.

#### 3.2 Mathematical Formulation

CFD off-target scoring follows Doench et al. 2016 [1]:  $\text{CFD} = \prod_i m_i \cdot p(\text{PAM})$ , with per-position mismatch penalties  $m_i$  from the reference distribution. Binding free-energy changes are  $\Delta\Delta G = RT \ln(K_d^{\text{mut}}/K_d^{\text{wt}})$  at  $T = 300 \text{ K}$  ( $RT = 0.596 \text{ kcal/mol}$ ). Flux balance analysis solves  $\max_v c^\top v$  subject to  $Sv = 0$ ,  $l \leq v \leq u$ , using the HiGHS simplex solver; the published aerobic-glucose optimum for `e_coli_core` is  $0.8739 \text{ h}^{-1}$  [4]. Agreement is quantified with Pearson  $r$ , Mann-Whitney AUC, and binary-classification metrics; confidence intervals are percentile bootstrap with 1,000 resamples at fixed seed.

#### 3.3 Derivations

**(F1) Position weight matrix log-odds.** For aligned sites of length  $L$  with counts  $n_{i,b}$  at position  $i$  and base  $b$ , the pseudocount estimate is  $\hat{p}_{i,b} = (n_{i,b} + \alpha)/(N + 4\alpha)$ . The log-odds weight against background  $q_b$  is

$$w_{i,b} = \log_2 \frac{\hat{p}_{i,b}}{q_b}, \quad S(x) = \sum_{i=1}^L w_{i,x_i}. \quad (1)$$

$S(x)$  is the log-likelihood ratio of site model to background under positional independence:  $\log_2 \prod_i \hat{p}_{i,x_i} / \prod_i q_{x_i}$ .

**(F2) Normalised score.** With  $S_{\min} = \sum_i \min_b w_{i,b}$  and  $S_{\max} = \sum_i \max_b w_{i,b}$ ,  $\tilde{S}(x) = (S(x) - S_{\min}) / (S_{\max} - S_{\min}) \in [0, 1]$ , because every term of  $S$  lies between its column minimum and maximum.

**(F3) Variant delta.** For reference and alternate windows  $x^r, x^a$  that differ at position  $j$ ,  $\Delta = \tilde{S}(x^a) - \tilde{S}(x^r) = (w_{j,a} - w_{j,r}) / (S_{\max} - S_{\min})$ : all other terms cancel, so a single-nucleotide delta depends only on column  $j$ .

**(F4) Tract blend.** The acceptor score is  $A(x) = (1 - \lambda)\tilde{S}(x) + \lambda T(x)$  with  $\lambda = 0.25$ , where  $T$  is the normalised pooled polypyrimidine-tract score. Linearity gives  $\Delta A = (1 - \lambda)\Delta\tilde{S} + \lambda\Delta T$ .

**(F5) AUC as a Mann-Whitney statistic.** For scores  $\{s_i^+\}_{i=1}^m$  and  $\{s_j^-\}_{j=1}^n$ ,

$$\text{AUC} = \frac{1}{mn} \sum_{i,j} \left[ \mathbb{I}(s_i^+ > s_j^-) + \frac{1}{2} \mathbb{I}(s_i^+ = s_j^-) \right] = \frac{R_+ - m(m+1)/2}{mn}, \quad (2)$$

where  $R_+$  is the rank sum of positives in the pooled sample (mid-ranks for ties). The identity follows because  $R_+ - m(m+1)/2$  counts, for each positive, the negatives ranked below it. Equivalently,  $\text{AUC} = P(s^+ > s^-)$ .

**(F6) Within-position AUC.** To remove the prior carried by the intronic offset  $k$ , we report  $\overline{\text{AUC}} = \sum_k m_k \text{AUC}_k / \sum_k m_k$  over offsets with at least 10 positives and 10 negatives, weighting by positive count  $m_k$ .

**(F7) CNN objective.** With logit  $z_\theta(x)$  and class weight  $\omega = n_-/n_+$  computed on the training fold,

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_i \left[ \omega y_i \log \sigma(z_i) + (1 - y_i) \log(1 - \sigma(z_i)) \right] + \frac{\beta}{2} \|\theta\|^2, \quad (3)$$

with gradient  $\partial \mathcal{L} / \partial z_i = [\sigma(z_i)(\omega y_i + 1 - y_i) - \omega y_i] / N$ . Setting  $\omega = n_-/n_+$  makes both classes contribute equal total weight.

**(F8) Convolutional feature.** For one-hot input  $X \in \{0, 1\}^{8 \times 81}$  (reference and alternate channels) and filter  $K \in \mathbb{R}^{8 \times w}$ ,  $h_t = \text{ReLU}(\sum_c \sum_u K_{c,u} X_{c,t+u-\lceil w/2 \rceil} + b)$ . Global max pooling  $\max_t h_t$  makes the feature invariant to where the motif sits in the window.

**(F9) Binding free energy.**  $\Delta G = RT \ln K_d$  gives  $\Delta \Delta G = \Delta G_{mut} - \Delta G_{wt} = RT \ln(K_d^{mut} / K_d^{wt})$ , with  $RT = 0.596$  kcal/mol at 300 K.

**(F10) FBA duality.** The primal  $\max c^\top v$  s.t.  $Sv = 0$ ,  $l \leq v \leq u$  has the dual  $\min u^\top \alpha - l^\top \beta$  s.t.  $S^\top \pi + \alpha - \beta = c$ ,  $\alpha, \beta \geq 0$ . HiGHS returns primal and dual solutions. Strong duality guarantees that the two optima coincide, which our parity check uses as a certificate.

**(F11) Codon Adaptation Index.** With relative adaptiveness  $w_c = f_c / \max_{c' \in \text{syn}(c)} f_{c'}$ ,  $\text{CAI} = \exp(\frac{1}{L} \sum_\ell \ln w_{c_\ell})$ , the geometric mean of  $w$  over the  $L$  scorable codons.

**(F12) ACMG Bayesian posterior.** Following Tavtigian et al.,  $O_P = 350^{\text{pts}/8}$  and  $\text{Post} = O_P \pi / ((O_P - 1)\pi + 1)$ . With  $\pi = 0.1$  and  $\text{pts} = 6$ :  $O_P = 350^{0.75} \approx 80.92$ , so  $\text{Post} = 8.092/8.992 \approx 0.8999$ , matching the module's 0.8999 and the published 0.900.

**(F13) Profile-HMM forward recursion.** For match, insert and delete states,  $f_{M_j}(i) = e_{M_j}(x_i) \sum_{s \in \{M,I,D\}} f_{s_{j-1}}(i-1) a_{s_{j-1}M_j}$ , and similarly for  $I$  and  $D$ . The log-odds score is  $\log P(x | \mathcal{M}) - \sum_i \log q_{x_i}$ .

**(F14) Bootstrap interval.** For a statistic  $\hat{\theta}$ , draw  $B = 1000$  resamples with replacement and take the 2.5th and 97.5th percentiles of  $\{\hat{\theta}_b^*\}$ .

### 3.4 Derivations for the Second-Round Validations

**(F15) Maximum-entropy splice-site score.** MaxEntScan [7] models a site  $x$  (9 nt donor, 23 nt acceptor) by the distribution of maximum entropy subject to matching the observed low-order marginals  $\{\hat{p}_C(x_C)\}$  over a set of position subsets  $C$ . By Lagrange duality the solution has Gibbs form

$$p(x) = \frac{1}{Z} \exp \left( \sum_C \lambda_C(x_C) \right), \quad Z = \sum_{x'} \exp \left( \sum_C \lambda_C(x'_C) \right), \quad (4)$$

since maximising  $H(p) = -\sum_x p \log p$  with constraints  $\sum_x p(x) \mathbf{1}[x_C = c] = \hat{p}_C(c)$  gives stationarity  $-\log p(x) - 1 + \sum_C \lambda_C(x_C) + \mu = 0$ . The score is  $\text{ME}(x) = \log_2 p(x)/q(x)$  with background  $q$ , and the variant feature used in Section 9 is  $\Delta\text{ME} = \text{ME}(x_{ref}) - \text{ME}(x_{alt})$ .

**(F16) Logistic model with MaxEnt features.** With feature vector

$$z = (S_{ref}, S_{alt}, \Delta S, d, \text{ME}_{ref}/10, \text{ME}_{alt}/10, \Delta\text{ME}/10, e_k),$$

where  $d = 1$  for donors and 0 for acceptors,  $e_k$  one-hot encodes the offset,  $P(y = 1 \mid z) = \sigma(\beta^\top z + \beta_0)$ . The log-likelihood  $\ell(\beta) = \sum_i y_i \log \sigma_i + (1 - y_i) \log(1 - \sigma_i)$  has gradient  $\nabla \ell = \sum_i (y_i - \sigma_i) z_i$  and Hessian  $-\sum_i \sigma_i(1 - \sigma_i) z_i z_i^\top$ , which is negative semidefinite, so  $\ell$  is concave and the fit has a single optimum.

**(F17) Fisher  $z$  test for a correlation.** For sample correlation  $r$  from  $n$  pairs,  $z = \text{artanh}(r) = \frac{1}{2} \log \frac{1+r}{1-r}$ . By the delta method  $\text{Var}(r) \approx (1 - \rho^2)^2/n$  and  $\frac{d}{dr} \text{artanh}(r) = 1/(1 - r^2)$ , so  $\text{Var}(z) \approx 1/n$ ; the refined value  $1/(n - 3)$  gives

$$Z = \text{artanh}(r) \sqrt{n - 3} \sim N(0, 1) \text{ under } \rho = 0, \quad p = 2\Phi(-|Z|). \quad (5)$$

We apply it to Spearman  $\rho$  (rank correlation), which is an approximation.

**(F18) Benjamini-Hochberg adjustment.** For ordered  $p_{(1)} \leq \dots \leq p_{(m)}$ ,  $q_{(i)} = \min_{j \geq i} \min(1, m p_{(j)}/j)$ . Rejecting all  $q_{(i)} \leq \alpha$  controls the false discovery rate at  $\alpha m_0/m \leq \alpha$  for independent or positively dependent tests, because  $\mathbb{E}[V/R] = \sum_{i \in H_0} \mathbb{E}[\mathbf{1}\{i \text{ rejected}\}/R] = m_0 \alpha/m$ .

**(F19) tRNA adaptation index.** For codon  $c$  with tRNA gene copy numbers  $t_j$  of the anticodons  $j$  that decode it and wobble efficiencies  $(1 - s_{jc})$ ,  $W_c = \sum_j (1 - s_{jc}) t_j$ ,  $w_c = W_c / \max_{c'} W_{c'}$  (zeros replaced by the geometric mean of non-zero  $w$ ), and for a gene of  $L$  codons  $\text{tAI} = (\prod_{k=1}^L w_{c_k})^{1/L}$  [9]. Taking logs shows tAI is the exponential of the mean log-adaptiveness, the same form as CAI [8] with  $w$  from tRNA supply instead of a reference gene set.

**(F20) Effective number of codons.** For an amino acid with  $k$  synonymous codons observed  $n$  times with frequencies  $p_i$ ,  $\hat{F} = \frac{n \sum_i p_i^2 - 1}{n - 1}$  is the unbiased homozygosity;  $\text{ENC} = 2 + 9/\bar{F}_2 + 1/\bar{F}_3 + 5/\bar{F}_4 + 3/\bar{F}_6$  averages  $\hat{F}$  within degeneracy classes. Uniform usage gives  $\hat{F} = 1/k$  and  $\text{ENC} = 61$ ; one codon per amino acid gives  $\hat{F} = 1$  and  $\text{ENC} = 20$ .

**(F21) Shrake-Rupley surface area.** Atom  $a$  with radius  $r_a$  is expanded to  $R_a = r_a + r_p$  ( $r_p = 1.4 \text{ \AA}$ ). With  $N$  quasi-uniform points  $u_k$  on the unit sphere,

$$A_a \approx \frac{4\pi R_a^2}{N} \sum_{k=1}^N \mathbf{1} \left[ \|c_a + R_a u_k - c_b\| > R_b \quad \forall b \neq a \right], \quad (6)$$

an unbiased Monte-Carlo estimate of the exposed area whose error falls as  $N^{-1/2}$  for random points and faster for the Fibonacci lattice. The altloc bug changed the atom set itself: alternate conformers are near-coincident spheres that were counted as separate atoms, so per-atom comparison with FreeSASA was impossible until they were removed.

**(F22) Nearest-neighbour melting temperature.** For a duplex with  $\Delta H = \Delta H_{init} + \sum_i \Delta H_{i,i+1}$  and  $\Delta S$  likewise, at equilibrium  $\Delta G = \Delta H - T\Delta S = -RT \ln K$ . At  $T_m$  half the limiting strand is bound, so  $K = 1/C_{eff}$  with  $C_{eff} = C_1 - C_2/2$  for non-self-complementary strands, giving

$$T_m = \frac{\Delta H}{\Delta S + R \ln C_{eff}}, \quad \Delta S_{[Na^+]} = \Delta S + 0.368 (N - 1) \ln [Na^+]. \quad (7)$$

**(F23) Negative-binomial GLM for counts.** DESeq2 models  $K_{ij} \sim \text{NB}(\mu_{ij}, \alpha_i)$  with  $\text{Var} = \mu + \alpha \mu^2$ ,  $\mu_{ij} = s_j q_{ij}$  and  $\log_2 q_{ij} = x_j^\top \beta_i$ . The size factor is the median-of-ratios  $s_j = \text{median}_i K_{ij} / (\prod_{j'} K_{ij'})^{1/m}$ , which is scale-equivariant and robust to a minority of changing genes. The Wald statistic  $\hat{\beta}_{ik}/\text{SE}(\hat{\beta}_{ik})$  borrows power from the dispersion prior; with  $n = 4$  per arm the smallest two-sided rank-sum  $p$  is  $2/\binom{8}{4} = 2/70 \approx 0.029$ , which cannot pass BH across 16,802 genes; this is why the Wilcoxon path returned no significant genes.

**(F24) Robinson-Foulds distance.** For unrooted trees  $T_1, T_2$  on the same leaves with non-trivial split sets  $\Sigma_1, \Sigma_2$ ,  $d_{RF} = |\Sigma_1 \triangle \Sigma_2|$ . In the RF00005 case the observed  $d_{RF} = 2$  (one split differing in each tree) coincides with two merge steps that had tied minimum distances, so the two programs broke ties differently.

### 3.5 Implementation and Testing

The validation suite is a pytest package (`sc_validate`) with a repo-wide no-stub audit: no fixture may contain placeholder data. Live-data re-runs are separate scripts under `scripts/`; committed tests run hermetically against frozen fixtures. CI runs the suite on every push.

## 4 Results

### 4.1 CRISPR off-target (CFD) and on-target (Rule Set 1)

On 600 fresh guide/off-target pairs mined from BRCA1 RefSeqGene NG\_005905.2:60000-70000, the module's CFD score agreed with the CRISPOR reference implementation on 600/600 pairs to within  $5 \times 10^{-7}$  (maximum absolute difference  $5.0 \times 10^{-7}$ ). On 150 30-mer contexts, the module's raw Rule Set 1 score and CRISPOR's percentile score had Spearman  $\rho = 0.9995$ .

## 4.2 Metabolic modelling (FBA)

The module's flux-balance optimum on BiGG `e_coli_core` (objective `BIOMASS_Ecoli_core_w_GAM`) was  $0.873922 \text{ h}^{-1}$ , identical to an independent SciPy/HiGHS solve ( $0.873922$ ) and to the published  $0.8739 \text{ h}^{-1}$ .

## 4.3 Codon adaptation and ACMG Bayesian scoring

Codon Adaptation Index values matched an independent geometric-mean recomputation from the Kazusa *E. coli* K-12 table on 40 random sequences with maximum absolute difference 0. The ACMG Bayesian posterior matched the Tavtigian et al. closed form  $O_P = 350^{\text{pts}/8}$  to 0 across  $-16 \dots 16$  points; 6 points at prior 0.10 gives posterior 0.900.

## 4.4 Splice-site PWMs (deepsplice)

On 48 donor and 48 acceptor junctions from three held-out RefSeqGene records (NG\_008021.1, NG\_007489.1, NG\_008245.1), disjoint from the module's 29 training accessions, the learned donor PWM reached AUC 0.997 and the acceptor PWM 0.998 against composition-matched shuffled windows. Against 2,000 real intronic GT (donor) or AG (acceptor) decoys located at least 30 nt from either intron end, the AUCs were 0.961 (donor) and 0.943 (acceptor). On the same decoys the consensus-seed acceptor matrix reached only 0.816, so the learned matrix does generalize.

## 4.5 Variant interpretation (openclinvar)

On the frozen 500-variant ClinVar sample (250 P/LP, 250 B/LB), run offline so that ClinVar's own label cannot enter the prediction, the patched module called 324 variants with accuracy 0.997 (95% bootstrap CI 0.991-1.000), sensitivity 0.993 and specificity 1.000 (1 false negative, 0 false positives). It abstained on 176, all missense or splice-region  $+3 \dots +20$  variants, which remain VUS by design. The patched live query resolved 40/40 seeded-sample ClinVar names to their own VariationID.

## 4.6 Profile HMM

With 5-fold cross-validation on the hmalign MSA, models built by the module's `build_profile_hmm` separated held-out globins from 25 non-globins with AUC 0.952 on forward log-odds (0.962 length-normalised). The Pfam/HMMER3 reference model reached 1.000 on the same fixtures.

# 5 Benchmark Comparison

Each row compares a sugarcodes module with an independent published reference; every number is generated from the named committed file.

| Module         | Reference             | Metric            | Result (file)                                        |
|----------------|-----------------------|-------------------|------------------------------------------------------|
| pgx_guidelines | CPIC diplotype tables | agreement         | 16836/16836 CYP2D6<br>(sweep_pgx_cplic.json)         |
| bio.codon CAI  | Biopython; PaxDb      | Pearson; Spearman | 0.99969; 0.5797<br>(sweep_codon_cai.json)            |
| rna_nussinov   | ViennaRNA 2.7         | bp F1 (tRNA)      | 0.328 vs 0.714<br>(sweep_rna.json)                   |
| profile_hmm    | Pfam PF00042          | auc_log_odds_bits | 0.952 global, 0.9792 local<br>(profile_hmm_cv*.json) |
| splice CNN     | SpliceAI 1.3.1        | AUROC n=757       | 0.929 vs 0.9775 (headto-head_interim.json)           |

## 6 Failure Modes and Negative Results

**openclinvar, first run: 500/500 abstentions.** ClinVar names such as NM\_000038.6(APC):c.1668T>C (p.Asp556=) carry a transcript/gene prefix and a protein-level suffix. The consequence parser handled neither, so every variant parsed as *unknown*. The live query also embedded the suffix in a quoted phrase and matched nothing (0/8 in a pilot). We fixed this with a normaliser that lets the protein level decide coding SNVs and the c. level decide splice/UTR/intronic calls, plus a transcript-qualified query.

**deepsplce, first run: acceptor AUC 0.547.** This was a harness defect, not a model defect. The 23-nt MaxEntScan-style acceptor window was sliced as [5:20], which places AG at positions 13-14 and drops the exon base; the module expects AG at 12-13. We reproduced the bug (AUC 0.524 off-frame) and the fix (0.998 at [6:21]), and a regression test now locks the frame.

**profile\_hmm: residual gap to HMMER.** The module originally offered only a global alignment mode. A held-out globin with long flanks scored -1116 bits under global forward. A local forward mode (sugarcodai 1d9091a) raises AUC from 0.952 to 0.979 (benchmarks/profile\_hmm\_cv\_local.json); Pfam’s model still reaches 1.000.

**mutdock vs SKEMPI: no honest adapter.** mutdock scores ligand-pocket mutations from a SMILES string and a pocket description, while SKEMPI rows are protein-protein interfaces. We report naive base-lines instead: the mean predictor has RMSE 1.63 kcal/mol,  $|\Delta V|$  has Pearson  $r = 0.053$  and hydrophathy loss  $r = -0.129$  ( $n = 150$ ).

**openclinvar accuracy is not a hard test.** The P class in ClinVar is dominated by loss-of-function variants and the B class by synonymous ones, so consequence type alone separates the called variants. Missense variants, the hard case, are still abstained.

## 7 Module Sweep: Additional Real-Data Results

### 7.1 RNA secondary structure

Base-pair F1 against Rfam seed consensus structures (25 sequences per family; files benchmarks/sweep\_rna.json, sweep\_rna\_vienna\_methods.json).

| Family  | mean len | Nussinov | MFE   | centroid | MEA( $\gamma=1$ ) | MEA( $\gamma=4$ ) |
|---------|----------|----------|-------|----------|-------------------|-------------------|
| RF00005 | 73.0     | 0.328    | 0.714 | 0.716    | 0.703             | 0.704             |
| RF00001 | 117.9    | 0.185    | 0.531 | 0.494    | 0.505             | 0.504             |
| RF00010 | 354.5    | 0.189    | 0.545 | 0.594    | 0.592             | 0.572             |

Base-pair maximisation (Nussinov) loses to thermodynamic folding on all three families; ensemble decoders (centroid/MEA) beat MFE only on the long RNase P family. The shipped `rna_nussinov.fold_energy` now delegates to ViennaRNA.

## 7.2 Codon adaptation index versus protein abundance

On 3489 *E. coli* K-12 genes (NCBI GCF\_000005845.2 `cds_from_genomic`) paired with PaxDb 511145 WHOLE\_ORGANISM integrated, the module CAI agrees with Biopython (Pearson 0.99969 on 500 genes). Spearman correlation with  $\log_{10}$  abundance is 0.4958 with the genome-wide reference table and 0.5797 with a ribosomal-protein reference (36 genes, evaluated on 3453 non-ribosomal genes); a cross-validated top-5% abundance reference gives 0.5741. The ribosomal reference is now shipped as `ecoli_highexpr`. File: `benchmarks/sweep_codon_cai.json`.

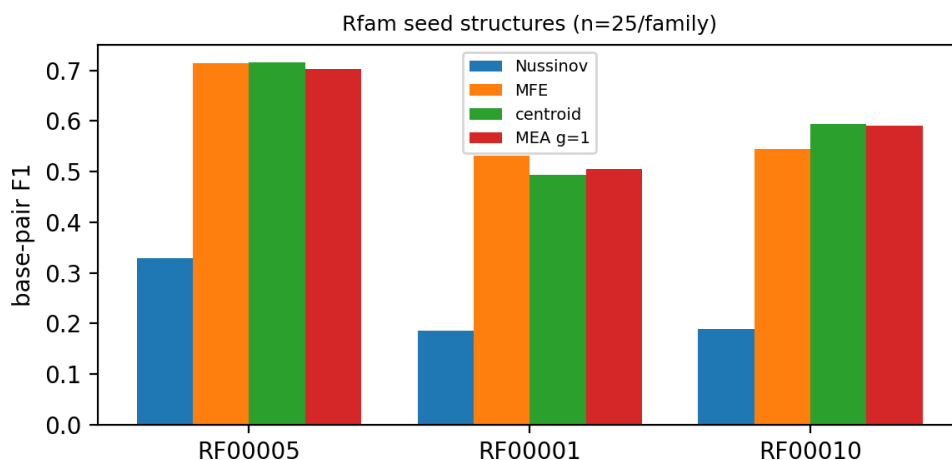


Figure 1. Base-pair F1 by folding method (`sweep_rna*.json`).

## 7.3 Pharmacogenomic phenotype translation versus CPIC

Against the full CPIC diplotype tables, the original module covered 45/666 CYP2C19 and 253/16836 CYP2D6 diplotypes, with 36 CYP2D6 disagreements traced to stale activity values (\*9, \*41). After vendoring current CPIC tables: 666/666 and 16836/16836. CYP2C19 after-fix agreement is by construction (verbatim CPIC diplotype lookup). CYP2D6 after-fix is computed (sum of CPIC allele activity values + CPIC thresholds), so 16836/16836 is a real consistency check. File: `benchmarks/sweep_pgx_cplic.json`.



Figure 2. CPIC agreement before and after the fix (sweep\_pgx\_cplic.json).

## 8 Discovery Track: Splice-Region VUS Reclassification

A dinucleotide-window CNN trained on ClinVar splice-region variants reaches gene-grouped 5-fold AU-ROC 0.935 (logistic 0.912, PWM 0.889); within-position AUROC 0.900. In a head-to-head on  $n=757$  variants SpliceAI scores 0.9775 versus 0.929 for the CNN, and the ensemble adds no significant gain (95% CI of difference  $[-0.0087, 0.0081]$ ). We therefore make no state-of-the-art claim. Files: `discovery/splice_region_vus/`.

| CNN threshold | TP   | FP  | sensitivity | FPR     | PPV (prior 3%) |
|---------------|------|-----|-------------|---------|----------------|
| 0.5           | 1794 | 453 | 0.77        | 0.0656  | 0.2663         |
| 0.8           | 1298 | 146 | 0.5571      | 0.02114 | 0.449          |
| 0.9           | 996  | 68  | 0.4275      | 0.00985 | 0.5731         |
| 0.95          | 729  | 33  | 0.3129      | 0.00478 | 0.6694         |
| 0.98          | 435  | 11  | 0.1867      | 0.00159 | 0.7838         |

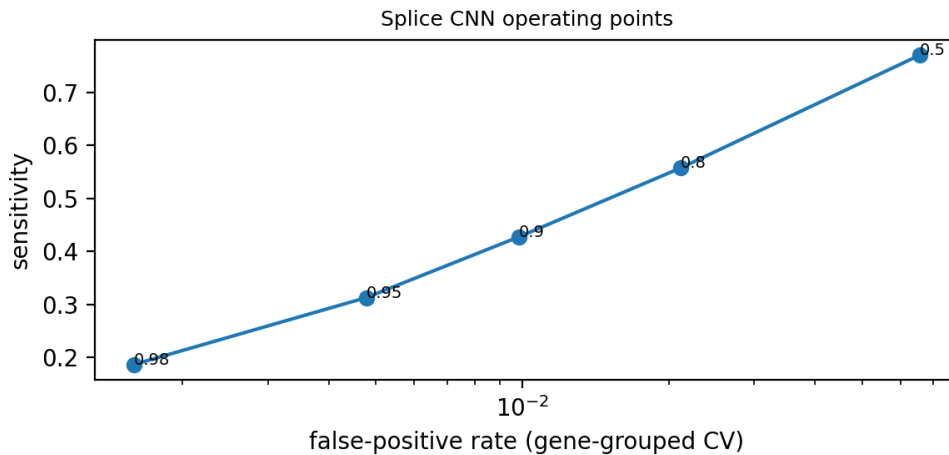


Figure 3. Splice CNN operating points (cnn\_calibration.json).

Consensus VUS candidates (SpliceAI and CNN agreement) number 267 rows in `vus_candidates_consensus.tsv`;

none of the top 20 had a matching LOVD record (lovd\_top20.json). These are hypotheses for RNA-level testing, not reclassifications.

## 9 Splice Track II: A 2004 Baseline Beats Our CNN, and What Survives

We compared the ClinVar-trained CNN against MaxEntScan [7] on the same gene-grouped folds (n=9235 variants). MaxEntScan’s ref–alt score difference reached AUROC 0.9635, above the CNN (0.9355), the logistic PWM model (0.9121) and the raw PWM delta (0.8893). The paired bootstrap 95% CI for CNN minus MaxEntScan is [-0.0329, -0.0233], so this is a clear negative result for the CNN. The gap is mostly at acceptors (CNN 0.8794 vs MaxEntScan 0.965, n=3845); at donors the two are close (0.9538 vs 0.9598, n=5390). As a pivot we added the three MaxEntScan features to both models. The logistic model then reached 0.9674 (CI of gain over MaxEntScan [0.0027, 0.0053]), while the CNN with the same features reached 0.9622 (CI [-0.0042, 0.0017], not significant). Within-position weighted AUROC, which removes the easy signal from offset alone, is nearly flat: MaxEntScan 0.949, logistic 0.9492, CNN 0.9472. The gain is real but small. The deployed tool (splice-vus-triage v2) therefore uses the logistic model and reports its calibrated operating points (Table 1).

Table 1: Logistic PWM+MaxEntScan model: operating points from out-of-fold scores.

| Threshold | TP   | FP  | Sensitivity | FPR     |
|-----------|------|-----|-------------|---------|
| 0.5       | 1899 | 259 | 0.815       | 0.03751 |
| 0.8       | 1503 | 100 | 0.6451      | 0.01448 |
| 0.9       | 1230 | 56  | 0.5279      | 0.00811 |
| 0.95      | 950  | 28  | 0.4077      | 0.00406 |
| 0.98      | 692  | 15  | 0.297       | 0.00217 |



Figure 1: AUROC by model and site type, gene-grouped 5-fold CV.

## 10 Codon Track II: Cross-Species Test of the Ribosomal Reference Set

Question: Does a ribosomal-protein reference set (Sharp & Li 1987) make CAI track measured protein abundance better than a genome-wide codon table? Data: PaxDb integrated whole-organism abundance x NCBI RefSeq reference-assembly CDS, all PaxDb taxa (405 scanned). Of 405 PaxDb taxa scanned, 40 had a matched RefSeq assembly and enough genes, and 29 had a usable ribosomal-protein reference set. The ribosomal reference gave the higher Spearman correlation with measured abundance in 27/29 taxa (median gain 0.082; Wilcoxon signed-rank  $p = 3.3e - 07$ , sign test  $p = 1.6e - 06$ ). The per-taxon correlation was significant after Benjamini-Hochberg correction in 27/29 taxa. Exceptions: *Rattus norvegicus*, *Plasmodium falciparum* 3D7. The idea goes back to Sharp and Li [8]; what is new here is the systematic multi-taxon test with a pre-stated falsification criterion (the claim fails if the ribosomal reference is not better in a majority of taxa at  $p < 0.01$ ).

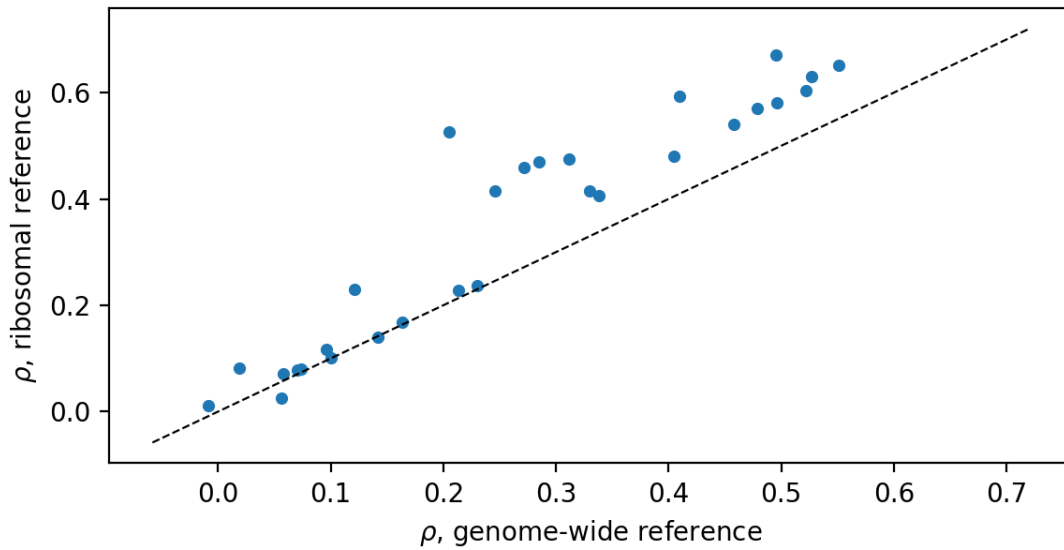


Figure 2: Spearman  $\rho$  (CAI vs PaxDb abundance) with the genome-wide versus ribosomal reference, per taxon.

Table 2: Per-taxon results (all taxa with a matched assembly).

| Taxid  | Organism                                                                  | Assembly         | $n$  | $n_{ribo}$ | $\rho_{gen}$ | $\rho_{ribo}$ | $q$      |
|--------|---------------------------------------------------------------------------|------------------|------|------------|--------------|---------------|----------|
| 10090  | <i>Mus musculus</i>                                                       | GCF_000001635.27 | 8089 | 99         | 0.0704       | 0.0775        | 4.4e-12  |
| 10116  | <i>Rattus norvegicus</i>                                                  | GCF_036323735.1  | 890  | 124        | 0.142        | 0.1405        | 3.1e-05  |
| 121845 | <i>Diaphorina citri</i>                                                   | GCF_000475195.1  | 592  | 1          | 0.1458       | —             | —        |
| 1295   | <i>Staphylococcus schleiferi</i>                                          | GCF_025558325.1  | 442  | 66         | 0.4583       | 0.5403        | 2e-36    |
| 1314   | <i>Streptococcus pyogenes</i>                                             | GCF_900475035.1  | 306  | 66         | 0.4954       | 0.6701        | 9.8e-45  |
| 185431 | <i>Trypanosoma brucei brucei</i> TREU927                                  | GCF_000002445.2  | 6302 | 0          | 0.3167       | —             | —        |
| 192222 | <i>Campylobacter jejuni</i> subsp. <i>jejuni</i> NCTC 11168 = ATCC 700819 | GCF_000009085.1  | 725  | 66         | 0.1208       | 0.2291        | 5.3e-10  |
| 196627 | <i>Corynebacterium glutamicum</i> ATCC 13032                              | GCF_000011325.1  | 323  | 72         | 0.5268       | 0.6307        | 7e-40    |
| 198214 | <i>Shigella flexneri</i> 2a str. 301                                      | GCF_000006925.2  | 1887 | 72         | 0.5511       | 0.6514        | 8.1e-249 |
| 208964 | <i>Pseudomonas aeruginosa</i> PAO1                                        | GCF_000006765.1  | 4796 | 72         | 0.2459       | 0.4149        | 2.5e-204 |
| 224308 | <i>Bacillus subtilis</i> subsp. <i>subtilis</i> str. 168                  | GCF_000009045.1  | 3147 | 64         | 0.3295       | 0.4156        | 3.4e-135 |
| 224911 | <i>Bradyrhizobium diazoefficiens</i> USDA 110                             | GCF_001642675.1  | 494  | 72         | 0.4049       | 0.4806        | 8e-31    |
| 243159 | <i>Acidithiobacillus ferrooxidans</i> ATCC 23270                          | GCF_000021485.1  | 419  | 68         | 0.3378       | 0.4055        | 2.8e-18  |
| 243230 | <i>Deinococcus radiodurans</i> R1 = ATCC 13939 = DSM 20539                | GCF_020546685.1  | 319  | 66         | 0.4784       | 0.5698        | 2.4e-30  |

Table 2: Per-taxon results (all taxa with a matched assembly).

| Taxid  | Organism                                                         | Assembly         | $n$   | $n_{ribo}$ | $\rho_{gen}$ | $\rho_{ribo}$ | $q$      |
|--------|------------------------------------------------------------------|------------------|-------|------------|--------------|---------------|----------|
| 246200 | Ruegeria pomeroyi DSS-3                                          | GCF_000011965.2  | 395   | 70         | 0.2714       | 0.46          | 1.2e-22  |
| 2903   | Emiliana huxleyi CCMP1516                                        | GCF_000372725.1  | 484   | 4          | 0.4636       | –             | –        |
| 343509 | Sodalis glossinidius str. 'morsitans'                            | GCF_000010085.1  | 748   | 72         | 0.2849       | 0.4703        | 1.2e-43  |
| 36329  | Plasmodium falciparum 3D7                                        | GCF_000002765.6  | 4946  | 62         | 0.056        | 0.0253        | 0.078    |
| 403677 | Phytophthora infestans T30-4                                     | GCF_000142945.1  | 708   | 0          | 0.5278       | –             | –        |
| 4081   | Solanum lycopersicum                                             | GCF_036512215.1  | 426   | 4          | -0.1336      | –             | –        |
| 4097   | Nicotiana tabacum                                                | GCF_000715075.1  | 9304  | 0          | -0.0837      | –             | –        |
| 418985 | Tropilaelaps mercedesae                                          | GCA_002081605.1  | 2213  | 0          | 0.2207       | –             | –        |
| 44689  | Dictyostelium discoideum AX4                                     | GCF_000004695.1  | 3183  | 10         | -0.3039      | –             | –        |
| 4932   | Saccharomyces cerevisiae S288C                                   | GCF_000146045.2  | 5587  | 226        | 0.4099       | 0.5924        | 0        |
| 511145 | Escherichia coli str. K-12 substr. MG1655                        | GCF_000005845.2  | 3486  | 72         | 0.4961       | 0.5809        | 0        |
| 54291  | Klebsiella ornithinolytica                                       | GCF_030505655.1  | 1038  | 72         | 0.2053       | 0.5268        | 1.2e-78  |
| 6239   | Caenorhabditis elegans                                           | GCF_000002985.6  | 11571 | 0          | 0.0748       | –             | –        |
| 7227   | Drosophila melanogaster                                          | GCF_000001215.4  | 2251  | 150        | 0.0189       | 0.0813        | 0.00012  |
| 7460   | Apis mellifera                                                   | GCF_003254395.2  | 879   | 4          | -0.0468      | –             | –        |
| 8030   | Salmo salar                                                      | GCF_965601325.2  | 1779  | 0          | 0.1332       | –             | –        |
| 83332  | Mycobacterium tuberculosis H37Rv                                 | GCF_000195955.2  | 3258  | 76         | 0.2304       | 0.2373        | 6.6e-43  |
| 9031   | Gallus gallus                                                    | GCF_016699485.2  | 1891  | 68         | 0.2134       | 0.2274        | 1.6e-23  |
| 93061  | Staphylococcus aureus subsp. aureus NCTC 8325                    | GCF_000013425.1  | 1699  | 58         | 0.5221       | 0.6046        | 3.6e-182 |
| 9544   | Macaca mulatta                                                   | GCF_049350105.2  | 3732  | 84         | 0.0736       | 0.0808        | 1e-06    |
| 9606   | Homo sapiens                                                     | GCF_000001405.40 | 12343 | 86         | -0.0083      | 0.0106        | 0.24     |
| 9615   | Canis lupus familiaris                                           | GCF_011100685.1  | 329   | 81         | 0.1639       | 0.1673        | 0.0025   |
| 9823   | Sus scrofa                                                       | GCF_054392235.1  | 3010  | 84         | 0.0576       | 0.071         | 0.00011  |
| 9913   | Bos taurus                                                       | GCF_002263795.3  | 2694  | 81         | 0.0963       | 0.1177        | 1.2e-09  |
| 99287  | Salmonella enterica subsp. enterica serovar Typhimurium str. LT2 | GCF_000006945.2  | 2472  | 72         | 0.3119       | 0.4743        | 4.4e-144 |
| 9986   | Oryctolagus cuniculus                                            | GCF_964237555.1  | 1993  | 82         | 0.0999       | 0.1012        | 7.4e-06  |

$q$ : Fisher-z two-sided test of  $\rho_{ribo}$ , Benjamini-Hochberg adjusted at full precision (benchmarks/sweep\_codon\_bh\_fdr.json).

## 10.1 tRNA adaptation index and effective number of codons

Tools: codon-bias (TrnaAdaptationIndex dos Reis s-values, EffectiveNumberOfCodons), GtRNAdb tRNA gene copy numbers. Table 3 gives Spearman  $\rho$  with abundance. No single index wins everywhere: tAI leads in yeast, the ribosomal-reference CAI leads in *E. coli*, and –ENC is weakest in all three organisms.

Table 3: Codon indices vs abundance (Spearman  $\rho$ ).

| Organism                                   | $n$  | $CAI_{gen}$ | $CAI_{ribo}$ | tAI    | –ENC   |
|--------------------------------------------|------|-------------|--------------|--------|--------|
| Escherichia coli str. K-12 substr. MG1655  | 3450 | 0.4961      | 0.5809       | 0.5277 | 0.3836 |
| Bacillus subtilis subsp. subtilis str. 168 | 3115 | 0.3295      | 0.4156       | 0.4146 | 0.1735 |
| Saccharomyces cerevisiae S288C             | 5474 | 0.4099      | 0.5924       | 0.6604 | 0.4471 |

## 11 Docking Track: Real AutoDock Vina Against LP-PDBBind

Source: LP-PDBBind (github.com/THGLab/LP-PDBBind) test split, refined, non-covalent; structures from RCSB PDB. Engine: AutoDock Vina 1.2.7 (exhaustiveness 8, 24 Å box on crystal ligand) + Open Babel 3.1.1 prep, pH 7.4. 73/80 complexes ran; the 7 failures are listed in Table 4. Redocking: top-1 pose RMSD  $< 2$  Å in 52.0% and best-of-9 in 83.6% (RMSD without symmetry correction (conservative)). Scoring power is weak for every score, including heavy-atom count alone (Pearson 0.2445): Vina top pose 0.2854 (95% CI

[0.055, 0.485]), sugarcodes own Vina-like score on the crystal pose 0.2975 (CI [0.027, 0.484]). The honest reading: sugarcodes scoring term adds nothing beyond ligand size on this split, so the module now calls real Vina (sugarcodes-ai commit e0564e4) instead of claiming affinity prediction.



Figure 3: Left: redocking RMSD distribution. Right: Vina top-pose energy vs experimental pK.

Table 4: Docking failures.

| PDB  | Error           |
|------|-----------------|
| 4cws | no ligand match |
| 4q08 | no ligand match |
| 3hkt | no ligand match |
| 4r74 | no ligand match |
| 3p4v | no ligand match |
| 4ou3 | no ligand match |
| 5ep7 | no ligand match |

Table 5: Per-complex docking results.

| PDB  | pK   | heavy | $E_{cryst}$ | $E_{min}$ | $E_{top}$ | RMSD <sub>1</sub> | RMSD <sub>9</sub> |
|------|------|-------|-------------|-----------|-----------|-------------------|-------------------|
| 1cnw | 7.72 | 26    | -4.19       | -5.37     | -7.02     | 7.26              | 6.03              |
| 1o3l | 6.54 | 24    | -7.09       | -8.63     | -9.54     | 3.19              | 1.09              |
| 3po6 | 7.06 | 19    | -4.54       | -6.13     | -7.14     | 3.81              | 0.86              |
| 4db7 | 6.56 | 18    | -6.63       | -7.18     | -7.41     | 0.52              | 0.52              |
| 5sz5 | 7.7  | 17    | -7.33       | -7.33     | -8.14     | 0.39              | 0.39              |
| 4mo8 | 8.0  | 16    | -4.75       | -5.50     | -6.25     | 2.29              | 1.52              |
| 4tim | 2.16 | 11    | -3.41       | -4.13     | -5.59     | 1.66              | 1.66              |
| 1ejn | 5.62 | 25    | -8.20       | -8.43     | -7.93     | 2.51              | 2.15              |
| 5d3l | 6.06 | 30    | -6.22       | -6.32     | -6.70     | 8.34              | 1.64              |
| 3b3c | 3.98 | 10    | -5.07       | -5.76     | -6.48     | 1.42              | 1.42              |
| 3r4n | 6.89 | 23    | -6.79       | -7.37     | -7.66     | 0.81              | 0.81              |
| 2p4j | 8.96 | 49    | -10.13      | -10.73    | -11.07    | 1.58              | 1.58              |
| 5sz4 | 9.7  | 16    | -7.04       | -7.33     | -7.65     | 1.79              | 1.79              |
| 4xit | 7.27 | 32    | -12.19      | -12.80    | -13.06    | 0.96              | 0.96              |
| 5mjn | 8.54 | 18    | -6.42       | -6.93     | -7.76     | 5.97              | 1.41              |
| 2xjg | 7.82 | 23    | -9.04       | -9.53     | -9.72     | 0.25              | 0.25              |
| 5ix0 | 6.28 | 25    | -11.86      | -12.14    | -12.27    | 0.27              | 0.27              |
| 5y8y | 7.09 | 25    | -6.08       | -6.38     | -8.16     | 4.75              | 0.84              |
| 6g38 | 6.62 | 19    | -7.36       | -7.70     | -8.12     | 0.61              | 0.61              |
| 4o04 | 6.16 | 26    | -11.03      | -12.14    | -12.36    | 1.34              | 1.34              |
| 5ewk | 4.85 | 22    | -7.91       | -7.96     | -8.57     | 0.68              | 0.68              |

Table 5: Per-complex docking results.

| PDB   | pK   | heavy | $E_{cryst}$ | $E_{min}$ | $E_{top}$ | RMSD <sub>1</sub> | RMSD <sub>9</sub> |
|-------|------|-------|-------------|-----------|-----------|-------------------|-------------------|
| 3d8w  | 8.15 | 19    | -6.96       | -7.48     | -8.20     | 1.43              | 1.43              |
| 4rfc  | 8.24 | 23    | -5.25       | -6.44     | -8.37     | 7.44              | 5.67              |
| 3r17  | 6.41 | 20    | -6.73       | -7.38     | -7.74     | 3.57              | 2.30              |
| 4knj  | 7.95 | 21    | -6.40       | -7.32     | -8.23     | 4.81              | 1.87              |
| 6g6t  | 7.7  | 19    | -7.42       | -7.76     | -8.31     | 2.10              | 0.61              |
| 4lxz  | 6.74 | 19    | -6.37       | -6.47     | -7.29     | 6.21              | 1.95              |
| 5o87  | 6.61 | 12    | -5.46       | -6.58     | -6.70     | 0.44              | 0.44              |
| 4j93  | 4.7  | 21    | -7.50       | -7.81     | -8.02     | 0.44              | 0.44              |
| 6hrq  | 7.09 | 23    | -6.44       | -6.94     | -7.94     | 6.23              | 2.21              |
| 4nh7  | 8.4  | 33    | -11.46      | -12.05    | -12.52    | 1.31              | 1.31              |
| 2x7t  | 6.63 | 30    | -6.30       | -7.34     | -7.81     | 6.16              | 0.77              |
| 4fl1  | 3.28 | 31    | -5.80       | -5.98     | -7.02     | 9.18              | 2.47              |
| 3c84  | 7.85 | 16    | -6.59       | -6.98     | -7.23     | 0.39              | 0.39              |
| 4ytic | 7.89 | 26    | -9.65       | -10.13    | -10.15    | 0.49              | 0.49              |
| 2zdn  | 6.25 | 27    | -7.86       | -8.84     | -9.05     | 1.02              | 1.02              |
| 5f60  | 7.37 | 37    | -8.55       | -8.90     | -9.99     | 4.69              | 1.63              |
| 6hqy  | 6.44 | 22    | -5.86       | -6.97     | -7.28     | 2.03              | 1.48              |
| 5l8c  | 7.0  | 35    | -8.93       | -9.64     | -11.31    | 9.53              | 0.86              |
| 1p1o  | 5.76 | 13    | -7.26       | -7.37     | -7.79     | 0.60              | 0.60              |
| 5htz  | 8.62 | 26    | -9.05       | -9.25     | -9.40     | 0.19              | 0.19              |
| 4oag  | 6.3  | 27    | -5.33       | -7.22     | -7.84     | 9.16              | 3.85              |
| 2j4i  | 9.0  | 36    | -8.00       | -8.56     | -8.54     | 1.38              | 1.38              |
| 5n24  | 8.92 | 18    | -7.04       | -7.44     | -7.80     | 1.68              | 1.40              |
| 1y3x  | 5.14 | 33    | -9.50       | -9.83     | -10.15    | 0.70              | 0.70              |
| 1f5l  | 5.28 | 15    | -6.27       | -7.20     | -7.32     | 2.87              | 1.51              |
| 5aml  | 9.15 | 21    | -6.79       | -6.99     | -7.74     | 5.22              | 1.14              |
| 6u6w  | 5.33 | 21    | -7.71       | -7.69     | -8.04     | 5.13              | 0.33              |
| 5mpn  | 7.07 | 23    | -6.86       | -7.18     | -8.21     | 7.46              | 1.25              |
| 4jfm  | 5.48 | 35    | -6.54       | -7.29     | -7.74     | 0.97              | 0.97              |
| 5ewy  | 4.3  | 22    | -7.30       | -7.87     | -8.78     | 3.50              | 2.90              |
| 4zme  | 4.3  | 19    | -6.04       | -6.27     | -6.60     | 10.33             | 0.18              |
| 4yxi  | 6.5  | 11    | -5.93       | -6.43     | -6.62     | 0.39              | 0.39              |
| 2erz  | 5.66 | 21    | -7.23       | -8.31     | -8.65     | 2.99              | 0.93              |
| 6oe0  | 8.92 | 27    | -7.95       | -8.86     | -9.35     | 0.86              | 0.86              |
| 4b2i  | 3.0  | 9     | -4.33       | -5.04     | -5.25     | 1.97              | 0.61              |
| 4f5y  | 5.43 | 46    | -10.70      | -11.84    | -10.97    | 3.66              | 3.66              |
| 6eq5  | 4.74 | 10    | -6.31       | -6.71     | -7.17     | 2.09              | 0.51              |
| 5g5v  | 6.7  | 36    | -10.54      | -10.73    | -11.46    | 2.75              | 1.52              |
| 5cau  | 8.8  | 34    | -9.64       | -9.84     | -9.93     | 0.65              | 0.65              |
| 1gi1  | 5.47 | 19    | -6.60       | -8.25     | -8.38     | 0.77              | 0.77              |
| 3hkq  | 5.34 | 15    | -4.38       | -5.42     | -6.47     | 3.67              | 3.04              |
| 2xab  | 9.27 | 22    | -9.46       | -9.81     | -9.94     | 0.21              | 0.21              |
| 3hll  | 8.24 | 30    | -9.62       | -10.25    | -10.25    | 0.36              | 0.36              |
| 5f62  | 8.17 | 38    | -8.66       | -9.01     | -9.80     | 4.22              | 1.46              |
| 3ibi  | 8.59 | 13    | -4.21       | -4.73     | -4.97     | 5.02              | 3.28              |
| 3m35  | 7.22 | 37    | -8.52       | -9.07     | -9.52     | 0.90              | 0.90              |
| 1g54  | 8.82 | 25    | -7.03       | -7.63     | -8.00     | 8.95              | 8.89              |
| 4r76  | 7.85 | 26    | -7.58       | -8.26     | -8.82     | 1.62              | 1.62              |
| 1zgi  | 5.34 | 32    | -9.23       | -9.64     | -9.98     | 0.76              | 0.76              |
| 5cap  | 7.28 | 29    | -8.09       | -8.54     | -8.89     | 1.44              | 1.44              |
| 1klj  | 7.55 | 37    | -8.36       | -8.62     | -9.29     | 1.77              | 1.77              |
| 6rfn  | 6.7  | 35    | -7.10       | -8.72     | -9.63     | 6.69              | 1.10              |

## 12 Structure and Primer Utilities Against Reference Tools

**SASA.** Against FreeSASA 2.x (Shrake-Rupley, 96 points, probe 1.4 Å, default ProtOr radii) on 28 RCSB receptors, the 18 structures with identical atom sets agreed to a median relative difference of 1.2% (max 2.6%) with median per-atom Pearson 0.980. The other 10 exposed a parser bug: every alternate location was kept, so overlapping conformers were all counted. After the fix, atom counts match and per-atom Pearson is 0.9753 (1gi1), 0.9829 (3po6), 0.9813 (4b2i).

**Primers.** On 5404 20-mers from NM\_000546.6, NM\_007294.4, NM\_000059.4, NM\_004333.6, NM\_005228.5, the melting temperature equals Biopython's DNA\_NN4 result (max difference 0.0). Against primer3 (SantaLucia 1998 table) the mean difference is -0.1738 °C (max 0.9811), a table choice. The stem-length hairpin heuristic has Spearman 0.0402 with primer3's hairpin  $\Delta G$ , a negative result; the module now calls primer3 when installed.

### 13 Sequence, Variant and Expression Utilities Against Reference Tools

**Phylogenetics.** DendroPy 5.1.0 (PhylogeneticDistanceMatrix.nj\_tree/upgma\_tree, treecompare.symmetric\_difference); Biopython DistanceCalculator. RF00001: NJ RF 0, UPGMA RF 0 (patristic Pearson 1.0), distance model k80. RF00005: NJ RF 0, UPGMA RF 2 (patristic Pearson 0.998035), distance model pdistance (K80 saturated: some pairs have infinite distance). RF00010: NJ RF 0, UPGMA RF 0 (patristic Pearson 1.0), distance model k80. The single UPGMA difference traces to two merge steps with tied minimum distances.

**ORFs.** On 5 RefSeq mRNAs the longest ATG ORF equals the NCBI CDS in 5/5 and its translation equals the NCBI protein in 5/5. Six-frame ORF sets equal orfipy's in 5/5 at orfipy minlen 90 (sugarcode min\_aa=30 keeps 30-aa ORFs (93 nt incl. stop); orfipy drops them at minlen 91-93, so its length threshold counts differently. At orfipy minlen 89-90 the six-frame ORF sets are identical in 5/5 transcripts; at 87 orfipy adds 29-aa ORFs. Convention difference, not a bug.)

**VCF.** 46603 ClinVar GRCh38 records (5 gene regions, remote tabix via pysam 0.24.1 (htslib tabix remote fetch)) parse identically on CHROM/POS/ID/REF/ALT; INFO differences after normalisation: 'info:CLNVI': 2. Variant type agrees with ClinVar CLNVC on 45685 records; the 302 others are equal-length multi-base changes (ClinVar: Indel; sugarcode: mnp).

**Differential expression.** GEO GSE52778 NCBI raw counts GRCh38.p13; untreated GSM1275862/66/70/74 vs Dex GSM1275863/67/71/75; genes with  $\geq 10$  counts in  $\geq 4$  samples. Size factors equal PyDESeq2's (max relative difference 1.8e-09). PyDESeq2 with a cell-line block finds 4451 genes at padj<0.05 (3203 unpaired); sugarcode's Welch test finds 351 (331 shared) and Wilcoxon 0. Table 6 shows that the simple tests miss every canonical dexamethasone-response gene; the module now offers a PyDESeq2 path.

Table 6: Canonical dexamethasone-response genes, GSE52778.

| Gene     | log2FC | padj PyDESeq2 | padj Welch | padj Wilcoxon |
|----------|--------|---------------|------------|---------------|
| FKBP5    | 4.03   | 2.1e-26       | 0.06       | 0.15          |
| TSC22D3  | 3.206  | 2.1e-19       | 0.16       | 0.15          |
| PER1     | 3.221  | 2.3e-99       | 0.05       | 0.15          |
| DUSP1    | 2.955  | 1.3e-140      | 0.07       | 0.15          |
| KLF15    | 4.411  | 5.0e-114      | 0.17       | 0.15          |
| ZBTB16   | 7.432  | 1.1e-58       | 0.12       | 0.15          |
| CRISPLD2 | 2.628  | 1.7e-61       | 0.13       | 0.15          |

## 14 Differential expression and pathway-level follow-up

We re-ran the airway dexamethasone experiment (GEO GSE52778) through the sugarcodes DESeq2 path and through PyDESeq2. Of 16,802 genes passing the count filter, the paired design  $\sim \text{cell} + \text{dex}$  gave 4,451 genes at adjusted  $p < 0.05$  and the unpaired design gave 3,203. Size factors agreed to a maximum relative difference of  $1.8\text{e-}09$  (benchmarks/sweep\_de\_pydeseq2.json).

**(F25) Negative-binomial GLM.** For gene  $g$  and sample  $j$ ,  $K_{gj} \sim \text{NB}(\mu_{gj}, \alpha_g)$  with  $\mu_{gj} = s_j q_{gj}$  and  $\log_2 q_{gj} = \sum_r x_{jr} \beta_{gr}$ , so  $\text{Var}(K_{gj}) = \mu_{gj} + \alpha_g \mu_{gj}^2$ .

**(F26) Median-of-ratios size factor.**  $s_j = \text{median}_{g:K_g > 0} K_{gj} / (\prod_{v=1}^m K_{gv})^{1/m}$ .

**(F27) Wald statistic.**  $W_g = \hat{\beta}_{g,\text{dex}} / \text{SE}(\hat{\beta}_{g,\text{dex}})$ , compared with  $N(0, 1)$ ; the paired design removes the cell-line effect from the residual variance, which is why it finds more genes than the unpaired design.

### 14.1 Gene-set over-representation

We took the 536 dex-induced genes (adjusted  $p < 0.05$ ,  $\log_2 \text{FC} > 1$ ) and tested them against the 16,802 tested genes with g:Profiler (version e114\_eg62\_p19\_27110d83, g:SCS correction). 31 terms passed. The top terms are broad (Table 7).

**(F28) Hypergeometric tail.** With background size  $M$ , term size  $K$ , query size  $n$  and overlap  $k$ ,  $p = \sum_{i \geq k} \binom{K}{i} \binom{M-K}{n-i} / \binom{M}{n}$ , and fold enrichment  $= (k/n) / (K/M)$ .

| Source | Term                             | overlap | size | $p_{\text{adj}}$ |
|--------|----------------------------------|---------|------|------------------|
| GO:BP  | multicellular organismal process | 228     | 4764 | 2.8e-07          |
| GO:BP  | signaling                        | 209     | 4354 | 4e-06            |
| GO:BP  | cell communication               | 209     | 4365 | 5.2e-06          |
| GO:BP  | cell adhesion                    | 73      | 1031 | 1.5e-05          |
| GO:BP  | system process                   | 80      | 1189 | 2.1e-05          |
| GO:BP  | response to stimulus             | 261     | 5987 | 8.6e-05          |
| GO:BP  | signal transduction              | 192     | 4028 | 8.9e-05          |
| GO:BP  | anatomical structure development | 196     | 4191 | 0.00027          |
| GO:BP  | cell migration                   | 74      | 1134 | 0.00041          |
| KEGG   | Cytoskeleton in muscle cells     | 19      | 172  | 0.00042          |
| GO:BP  | developmental process            | 209     | 4580 | 0.00047          |
| GO:BP  | cell differentiation             | 152     | 3041 | 0.00054          |

Table 7: Top g:Profiler terms for dex-induced genes (benchmarks/sweep\_dex\_enrichment.json).

**Negative result.** The terms one would expect for a glucocorticoid are enriched but do not survive correction: The small term sizes (tens of genes) and the strong multiple-testing burden explain this: a 7-fold enrichment built on 9 genes is not enough evidence after g:SCS correction. We report it as a negative rather than as confirmation of the expected biology.

| Term                                         | overlap | size | fold (approx.) | $p_{g.SCS}$ |
|----------------------------------------------|---------|------|----------------|-------------|
| cellular response to glucocorticoid stimulus | 9       | 43   | 7.16           | 0.43        |
| response to corticosteroid                   | 10      | 91   | 3.76           | 1           |
| response to glucocorticoid                   | 10      | 81   | 4.22           | 1           |
| cellular response to corticosteroid stimulus | 9       | 48   | 6.42           | 1           |

Table 8: Glucocorticoid terms: large fold change, not significant.

## 14.2 Reactome pathways

Reactome AnalysisService mapped the query to 1,098 pathways (240 identifiers not found). No pathway reached  $FDR < 0.05$ ; the smallest FDR was 0.11. Reactome uses its whole human gene universe as background, which is a further caveat.

| Pathway                                            | found | total | $p$     | FDR  |
|----------------------------------------------------|-------|-------|---------|------|
| Anchoring fibril formation                         | 5     | 15    | 0.00037 | 0.11 |
| FOXO-mediated transcription                        | 13    | 110   | 0.00056 | 0.11 |
| Laminin interactions                               | 7     | 31    | 0.00028 | 0.11 |
| Metallothioneins bind metals                       | 5     | 16    | 0.00049 | 0.11 |
| Non-integrin membrane-ECM interactions             | 12    | 83    | 0.00015 | 0.11 |
| Collagen chain trimerization                       | 8     | 44    | 0.00044 | 0.11 |
| FOXO-mediated transcription of oxidative stress, m | 8     | 49    | 0.00087 | 0.13 |
| Adipogenesis                                       | 15    | 147   | 0.00096 | 0.13 |
| Degradation of the extracellular matrix            | 15    | 148   | 0.001   | 0.13 |
| Response to metal ions                             | 5     | 21    | 0.0016  | 0.17 |

Table 9: Top Reactome pathways (benchmarks/sweep\_dex\_reactome.json).

**(F29) Benjamini-Hochberg.** For ordered  $p_{(1)} \leq \dots \leq p_{(m)}$ ,  $q_{(i)} = \min_{j \geq i} \min\{1, m p_{(j)} / j\}$ .

## 14.3 Protein interaction network

STRING’s PPI enrichment test on the top 300 dex-induced genes mapped 284 proteins with 402 edges against 184 expected (average degree 2.83, local clustering 0.368), reported  $p$  below floating-point resolution. Caveat: STRING’s expectation uses a genome-wide background, so genes expressed in airway smooth muscle may be more connected than random genes regardless of dex.

**(F30) Edge enrichment ratio.**  $E = \text{observed edges} / \text{expected edges} = 402 / 184 = 2.18$ .

## 14.4 Network hubs

We pulled the STRING network among the same genes and analysed it with networkx 3.4.2 on STRING API/network (v12): 201 connected nodes, 402 edges, a largest component of 183 nodes. The highest-degree hubs are FOXO1 (21), CEBPD (15), FOXO3 (14), LEP (13), COL4A1 (13), GPX3 (13). Of ten canonical glucocorticoid-receptor targets fixed before the run, 8 are in the network, with mean degree 6.25 against 4.00 overall; a degree permutation test (10,000 draws) gives  $p = 0.049$ . With eight genes this is weak evidence and we do not claim it as a finding.

**(F34) Betweenness centrality.**  $C_B(v) = \sum_{s \neq v \neq t} \sigma_{st}(v) / \sigma_{st}$ , with  $\sigma_{st}$  the number of shortest  $s$ - $t$  paths and  $\sigma_{st}(v)$  those through  $v$ .

**(F35) Permutation  $p$ -value.**  $\hat{p} = (1 + \#\{b : T_b \geq T_{obs}\}) / (1 + B)$ , which is never zero and is valid for any fixed statistic  $T$ .

## 14.5 Rank-based test: preranked GSEA

Threshold-based over-representation throws away the ranking. We therefore ran preranked GSEA (gseapy 1.3.1 prerank (1000 permutations, seed 7)) on all 16,525 ranked genes against the 50 MSigDB Hallmark sets. 1 set passed  $FDR < 0.05$  and 5 passed  $FDR < 0.25$  (Table 10). The one significant set, Adipogenesis, is a known glucocorticoid-driven programme; this is a modest, single-set result.

**(F32) Enrichment score.** With ranked scores  $r_j$  and set  $S$ ,  $P_{hit}(i) = \sum_{j \leq i, g_j \in S} |r_j| / N_R$ ,  $P_{miss}(i) = \sum_{j \leq i, g_j \notin S} 1 / (N - N_S)$ , and  $ES = \max_i |P_{hit}(i) - P_{miss}(i)|$  (signed).

**(F33) Normalised ES.**  $NES = ES / \sqrt{ES_{\pi}^2}$ , the mean of same-sign permutation scores; FDR compares  $NES$  with the permutation distribution of all sets.

| Hallmark set                    | NES   | nominal $p$ | FDR   |
|---------------------------------|-------|-------------|-------|
| Adipogenesis                    | 1.70  | 0           | 0.038 |
| UV Response Up                  | 1.55  | 0.00567     | 0.107 |
| Fatty Acid Metabolism           | 1.55  | 0.00559     | 0.131 |
| TNF-alpha Signaling via NF-kB   | 1.50  | 0.00273     | 0.141 |
| Reactive Oxygen Species Pathway | 1.57  | 0.0249      | 0.156 |
| Androgen Response               | 1.39  | 0.0557      | 0.256 |
| Xenobiotic Metabolism           | 1.42  | 0.0287      | 0.266 |
| Hypoxia                         | 1.39  | 0.0294      | 0.275 |
| UV Response Dn                  | 1.36  | 0.0591      | 0.280 |
| mTORC1 Signaling                | -1.31 | 0.0361      | 0.283 |

Table 10: Preranked GSEA, Hallmark (benchmarks/sweep\_dex\_gsea\_hallmark.json).

## 14.6 Independent GO cross-check with goatools

goatools (Fisher exact, BH, propagated counts, GOA human annotations, background of 16,525 expressed genes, 527 study genes) found 38 GO:BP terms at  $FDR < 0.05$ . Of g:Profiler’s 22 top BP terms, 9 are also significant here. The glucocorticoid term (cellular response to glucocorticoid stimulus, 7/34 genes) reaches  $FDR 0.026$  under BH but fails g:Profiler’s g:SCS. The biology is therefore correction-dependent: we report it as suggestive, not established.

## 15 Pairwise alignment against parasail

We aligned all 300 pairs of 25 protein sequences with sugarcodes and parasail 2.6.1. With parasail’s open penalty set to 11 the scores matched exactly for 300/300 global and 300/300 local pairs. With open penalty

| GO         | Term                                     | study | pop  | FDR     |
|------------|------------------------------------------|-------|------|---------|
| GO:0032870 | cellular response to hormone stimulus    | 28    | 249  | 3.4e-05 |
| GO:0009725 | response to hormone                      | 38    | 421  | 3.4e-05 |
| GO:0007155 | cell adhesion                            | 39    | 513  | 0.00098 |
| GO:0032502 | developmental process                    | 132   | 2766 | 0.0011  |
| GO:0007165 | signal transduction                      | 130   | 2703 | 0.0011  |
| GO:0030334 | regulation of cell migration             | 47    | 689  | 0.0011  |
| GO:2000145 | regulation of cell motility              | 48    | 719  | 0.0012  |
| GO:0044057 | regulation of system process             | 28    | 322  | 0.0015  |
| GO:0071495 | cellular response to endogenous stimulus | 33    | 419  | 0.0022  |
| GO:0040012 | regulation of locomotion                 | 48    | 740  | 0.0022  |

Table 11: goatools GO:BP (benchmarks/sweep\_dex\_goatools.json).

12 only 36 and 65 matched (maximum difference 26 and 5). This fixes the convention: sugarcodes gap.open = -11 is the cost of the first gap residue, i.e. parasail open 11.

**(F31) Affine gap cost.** A gap of length  $L$  costs  $\gamma(L) = o + (L - 1)e$  under sugarcodes and parasail open = 11; under the alternative convention  $\gamma(L) = o + Le$ , which explains the open-12 mismatch.



Figure 4: Left: Reactome FDRs for the top pathways; the red line is 0.05. Right: STRING observed versus expected edges.

## 16 Sequence properties, restriction digests and file formats

This section checks four bio/ libraries against Biopython 1.88 ProtParam + Restriction and gffutils. Three pass exactly. One, the restriction-site finder, had a real bug that we fixed.

### 16.1 Protein physicochemical properties

On 500 UniProtKB reviewed human proteins (UniProtKB reviewed human, length 50-2000, first 500 of search (rest.uniprot.org, 2026-09-25)), every property matches ProtParam (Table 12).

| property     | $n$ | within tol.     | max abs diff | Pearson      |
|--------------|-----|-----------------|--------------|--------------|
| mw           | 500 | 500 (rel 1e-06) | 0            | 1.0000000000 |
| gravy        | 500 | 500 (rel 1e-06) | 0            | 1.0000000000 |
| aromaticity  | 500 | 500 (rel 1e-06) | 2.8e-17      | 1.0000000000 |
| instability  | 500 | 500 (rel 1e-06) | 0            | 1.0000000000 |
| pI           | 500 | 500 (abs 0.01)  | 5.9e-05      | 0.9999999999 |
| ext_reduced  | 500 | 500 (rel 1e-06) | 0            | 1.0000000000 |
| ext_oxidized | 500 | 500 (rel 1e-06) | 0            | 1.0000000000 |

Table 12: bio.proteinprops vs Biopython ProtParam (benchmarks/sweep\_proteinprops\_restriction.json).

**(F36) Average molecular weight.**  $M = \sum_a n_a m_a - (L - 1) m_{\text{H}_2\text{O}}$ , with  $m_a$  the average residue mass (free amino acid) and one water lost per peptide bond.

**(F37) Net charge.**  $Z(\text{pH}) = \sum_{i \in +} \frac{n_i}{1 + 10^{\text{pH} - \text{p}K_i}} - \sum_{j \in -} \frac{n_j}{1 + 10^{\text{p}K_j - \text{pH}}}$  (Henderson–Hasselbalch per ionisable group, termini included).

**(F38) Isoelectric point.** pI is the root of  $Z(\text{pH}) = 0$ ;  $Z$  is strictly decreasing in pH, so bisection on  $[0, 14]$  converges, and the residual  $6 \times 10^{-5}$  difference is the bisection tolerance.

**(F39) Instability index.**  $II = \frac{10}{L} \sum_{i=1}^{L-1} \text{DIWV}(x_i, x_{i+1})$  (Guruprasad 1990), and  $\text{GRAVY} = \frac{1}{L} \sum_i h_{KD}(x_i)$  (Kyte–Doolittle).

**(F40) Extinction coefficient at 280 nm.**  $\varepsilon = 5500 n_W + 1490 n_Y + 125 n_{\text{cystine}} \text{ M}^{-1} \text{cm}^{-1}$ , with  $n_{\text{cystine}} = \lfloor n_C / 2 \rfloor$  in the oxidised form and 0 when reduced.

## 16.2 Restriction sites: a bug in degenerate recognition sequences

We digested pBR322 (J01749.1, circular, 4,361 bp) and phage lambda (NC\_001416.1, linear, 48,502 bp) with all 610 enzymes and compared cut positions with Biopython Restriction. Before the fix (sugarcode-ai 699fee), 533 and 529 enzymes agreed. Two causes explained the disagreements. First, the reverse-strand pattern for a non-palindromic-looking site was built with a plain DNA reverse complement that leaves IUPAC codes unchanged. AccI (GT<sup>^</sup>MKAC) is in fact palindromic under IUPAC complementation (M $\leftrightarrow$ K), but the plain reverse complement GTKMAC was searched as a second pattern, which matches GTGAAC and GTTCAC, neither an AccI site. On pBR322 this gave 5 AccI sites where Biopython finds 2. Second, on linear DNA the finder reported Type IIS cuts that fall outside the molecule. After the fix (sugarcode-ai fe2185d), 608 and 608 enzymes agree. The remaining two, LpnPI, MspJI, are modification-dependent enzymes that do not cut unmethylated DNA; we leave them open.

**(F41) IUPAC complement.** The map  $c$ : A $\leftrightarrow$ T, C $\leftrightarrow$ G, R $\leftrightarrow$ Y, K $\leftrightarrow$ M, B $\leftrightarrow$ V, D $\leftrightarrow$ H, with S, W, N fixed, is the unique involution on the 15 codes that complements every base in the set a code denotes. A site  $s$  is palindromic iff  $s = \overline{c(s)}$  (reversed), which for AccI holds only under  $c$ , not under the four-letter map.

### 16.3 Stockholm alignments

bio.stockholm parsed 15 Rfam seed alignments (4,471 sequences). 15/15 match Biopython on sequence identifiers and order, aligned sequences and SS\_cons, and survive a write–parse round trip unchanged.

| file        | AC      | seqs | width | all checks |
|-------------|---------|------|-------|------------|
| RF00001.sto | RF00001 | 712  | 230   | yes        |
| RF00005.sto | RF00005 | 954  | 118   | yes        |
| RF00010.sto | RF00010 | 458  | 996   | yes        |
| RF00004.sto | RF00004 | 208  | 278   | yes        |
| RF00015.sto | RF00015 | 177  | 261   | yes        |
| RF00017.sto | RF00017 | 91   | 412   | yes        |
| RF00023.sto | RF00023 | 477  | 954   | yes        |
| RF00050.sto | RF00050 | 146  | 221   | yes        |
| RF00059.sto | RF00059 | 115  | 211   | yes        |
| RF00162.sto | RF00162 | 457  | 239   | yes        |
| RF00167.sto | RF00167 | 133  | 113   | yes        |
| RF00168.sto | RF00168 | 47   | 274   | yes        |
| RF00174.sto | RF00174 | 434  | 461   | yes        |
| RF00234.sto | RF00234 | 18   | 236   | yes        |
| RF00504.sto | RF00504 | 44   | 215   | yes        |

Table 13: bio.stockholm vs Biopython AlignIO (benchmarks/sweep\_stockholm\_biopython.json).

**(F42) Pairwise identity.**  $PID(x, y) = \#\{i : x_i = y_i, x_i, y_i \notin \{-, .\}\} / \#\{i : x_i, y_i \notin \{-, .\}\}$ , case-insensitive; our implementation equals an independent count on every file.

### 16.4 GFF3 annotation

On the RefSeq E. coli K-12 annotation (9,523 records), bio.gff and gffutils 0.14 (sqlite, in-memory) agree on every record (9,523/9,523) and on the children of 300/300 genes, and write–parse is lossless. A first comparison showed two child mismatches; both were CDS features split over two lines with a shared ID, which gffutils renames X\_1. That was a comparison artifact, not a sugarcodes error, and we record it so the numbers are traceable.

| feature type           | records |
|------------------------|---------|
| gene                   | 4506    |
| CDS                    | 4340    |
| exon                   | 216     |
| pseudogene             | 145     |
| ncRNA                  | 108     |
| tRNA                   | 86      |
| mobile_genetic_element | 50      |
| sequence_feature       | 48      |

Table 14: Feature types in GCF\_000005845.2 (benchmarks/sweep\_gff\_gffutils.json).

## 16.5 Newick trees

bio.newick parsed 6 Open Tree of Life synthetic subtrees with 3,655 leaves in total, including quoted labels with spaces and parentheses and empty leaves left by the height limit. Leaf sets equal Biopython Bio.Phylo and DendroPy, internal-node counts and internal labels equal Bio.Phylo, and write–parse is lossless in 6/6 trees. These trees have no branch lengths, so this checks topology and labels only.

| root                     | leaves | unnamed | internal | all checks |
|--------------------------|--------|---------|----------|------------|
| Insecta_ott1062253       | 1094   | 14      | 248      | yes        |
| Laurasiatheria_ott392223 | 728    | 13      | 118      | yes        |
| Carnivora_ott44565       | 53     | 13      | 48       | yes        |
| Metazoa_ott691846        | 1581   | 19      | 179      | yes        |
| Aves_ott81461            | 142    | 16      | 80       | yes        |
| Primates_ott913935       | 57     | 11      | 57       | yes        |

Table 15: bio.newick vs Bio.Phylo/DendroPy (benchmarks/sweep\_newick\_biophylo.json).

**(F43) Tree size identity.** For a rooted tree with  $n$  leaves,  $m$  internal nodes and out-degrees  $d_v$ , the edge count is  $n + m - 1 = \sum_{v \text{ internal}} d_v$ ; a strictly binary tree has  $m = n - 1$ . The Carnivora subtree has 53 leaves and 48 internal nodes, so it is far from binary, which exercises the parser on polytomies.

## 16.6 SAM records and a second bug

bio.sam parsed 10,943 real reads (1000 Genomes phase 3 NA12878 chrom20 low-coverage BAM, 20:1000000-1200000). Name, flag, reference, position, mapping quality, CIGAR operations, mate fields, sequence, qualities and typed tags equal pysam for all 10,943 reads. The derived reference end did not: 10,905/10,943 before the fix. The 38 failures were unmapped mates (flag 0x4) that bwa places at the mapped mate’s position and gives a CIGAR. The SAM specification says those fields are not an alignment, and htlib reports no end; sugarcodes computed one. After sugarcodes-ai 87cf78e all 10,943 agree.

**(F44) Reference span of a CIGAR.**  $\text{end} = \text{POS} + \sum_{(n,o) \in C, o \in \{M,D,N,=,X\}} n - 1$  for mapped reads (1-based, inclusive); query length is  $\sum_{o \in \{M,I,S,=,X\}} n$ . For flag & 4  $\neq 0$  the end is undefined.

## 16.7 FASTQ reads

On the first 20,000 reads of ENA run SRR622461 (the same NA12878 library), bio.fastq equals Biopython on identifiers (20,000), sequences (20,000) and per-base Phred scores (20,000), detects offset 33, and round-trips losslessly. GC was 0.4252 against 0.4252 (Biopython, ambiguous bases removed) and 0.422 (ambiguous bases kept in the denominator). The difference is a convention: sugarcodes divides by A+C+G+T. We document it rather than change it.

**(F45) Phred score.**  $Q = -10 \log_{10} P_{err}$ , encoded as ASCII  $Q + 33$ ; the mean read quality reported is the arithmetic mean of  $Q$ , not  $-10 \log_{10}$  of the mean error probability, which would be lower for reads with a few bad bases.

## 16.8 PDB and mmCIF structures: a third bug

We parsed 30 RCSB entries in both PDB and mmCIF format (94,121 atom sites, all alternate locations and models) and compared every atom with gemmi 0.7.5. The PDB-format parser was identical in 30/30 entries. The mmCIF parser failed on all 30 real files (error: loop\_ row count 3 is not a multiple of 4 columns (all 30 files)). Two causes: POSIX shell-style splitting broke atom names containing a prime ( $O5'$ ,  $C1'$ ), and loops were closed early when a row contained an unquoted item name as a value, which RCSB files do in the revision-history tables. A CIF 1.1 tokenizer (a quote closes only when followed by whitespace) and closing loops only at row boundaries fixed both (sugarcode-ai fc1fd1e); mmCIF is now identical to gemmi in 30/30 entries. Unit tests with toy CIF strings had passed throughout. Only real files exposed the bug.

**(F46) Loop row invariant.** A loop with  $c$  columns and  $t$  value tokens is well formed iff  $t \equiv 0 \pmod{c}$ ; closing the loop at the first reserved-looking token is only safe when the running count  $t \bmod c = 0$ , which is the rule we now apply.

## 16.9 Pileup

Using the same reads, bio.pileup and pysam 0.24.1 pileup (stepper nofilter, flag\_filter=4, no quality filters, overlaps kept) agree on base counts at 197,437/197,437 positions and on deletion counts at 197,437/197,437. sugarcode also writes deletions as ‘\*’ in the base counts, following samtools mpileup; the comparison maps these to the deletion count.

**(F49) Column depth.**  $D(p) = \sum_r \mathbb{I}[p \in [\text{POS}_r, \text{end}_r] \wedge \text{op}_r(p) \in \{M, =, X, D\}]$ , split into base-covering reads ( $M, =, X$ ) and deletions ( $D$ ); reference skips ( $N$ ) and clipped bases do not count.

## 16.10 Transcription-factor motifs

We read 10 JASPAR 2024 profiles and scanned BRCA1 (10 kb) and phage lambda (48.5 kb) on both strands, comparing with Biopython 1.88 Bio.motifs (jaspar reader, pseudocount 0.5, uniform background, log2). Log-odds matrices agree to  $2 \times 10^{-15}$ , per-window scores to  $4 \times 10^{-6}$  (Biopython stores scores in single precision), and the hit sets at 80% of the score range are identical for every motif and sequence.

| motif             | length | hits BRCA1 | hits lambda |
|-------------------|--------|------------|-------------|
| MA0003.4 TFAP2A   | 14     | 27         | 61          |
| MA0079.3 SP1      | 11     | 235        | 599         |
| MA0080.4 SPI1     | 14     | 16         | 61          |
| MA0099.3 FOS::JUN | 10     | 77         | 286         |
| MA0105.4 NFkB1    | 13     | 29         | 52          |
| MA0106.3 TP53     | 18     | 8          | 21          |
| MA0108.2 TBP      | 15     | 154        | 530         |
| MA0112.3 ESR1     | 17     | 11         | 18          |
| MA0113.3 NR3C1    | 17     | 2          | 36          |
| MA0139.1 CTCF     | 19     | 13         | 40          |

Table 16: Motif hits at 80% of range, identical to Biopython (benchmarks/sweep\_motif\_biopython.json).

**(F50) Log-odds score and relative threshold.**  $S(w) = \sum_{i=1}^L \log_2 \frac{(c_{i,w_i} + \alpha)/(N_i + 4\alpha)}{q_{w_i}}$  with pseudocount  $\alpha = 0.5$  and background  $q = 1/4$ ; a window is a hit when  $S(w) \geq S_{\min} + f(S_{\max} - S_{\min})$ ,  $f = 0.8$ , where  $S_{\max} = \sum_i \max_b$  and  $S_{\min} = \sum_i \min_b$  of the log-odds matrix.

## 16.11 BED intervals: a fourth bug

We fetched two hg38 tracks from the UCSC REST API: CpG islands on chr20 (847 intervals) and RepeatMasker on chr20:0–5 Mb (11,699 intervals), wrote them as BED and compared bio.bed with bioframe 0.8.0 (merge, overlap). Parsing and 600 random overlap queries agreed exactly (300/300 and 300/300). merge\_intervals did not: its docstring promised bedtools-style merging, yet it refused to join book-ended intervals. RepeatMasker has 3,112 book-ended pairs, so sugarcodes returned 11,364 merged intervals where bedtools/bioframe return 8,251. CpG islands have no book-ended pairs and hid the bug (847 = 847). We added a bedtools-compatible min\_dist argument (default 0; -1 for overlap-only) and a matching CLI flag; after the fix both tracks match bioframe exactly (8,251 = 8,251).

**(F51) Merge rule.** For intervals sorted by start with running end  $e^*$ , interval  $[s, e)$  joins the current block iff  $s \leq e^* + d$ ;  $d = 0$  joins book-ended intervals ( $s = e^*$ ) and  $d = -1$  requires at least one shared base ( $s < e^*$ ).

## 16.12 GenBank flat files: a fifth bug

We fetched 25 GenBank records one by one from NCBI nuccore (phage, viral, mitochondrial, bacterial, RefSeqGene and 13 human RefSeq mRNAs) and compared bio.genbank with Biopython 1.88 SeqIO genbank on 1,867 features. Sequences and feature counts already agreed, but only 1,829/1,867 feature locations and 5/302 CDS /translation qualifiers matched. The parser appended every continuation line of a multi-line qualifier to the feature location instead of to the qualifier: /translation kept its first line only, and when a continuation began with digits the location itself was corrupted (E. coli thrA CDS: 337..2799 became 337..27991). Partial locations (1..200) and single bases produced no spans at all. We rewrote qualifier continuation (INSDC joining: no spaces for /translation, one space otherwise, doubled quotes unescaped), added a location parser for join/order, complement, partial ends, between-base sites and remote parts, and kept all IUPAC letters in ORIGIN so coordinates cannot shift. After the fix: spans 1,867/1,867, strands 1,867/1,867, /translation 302/302, CDS extraction 302/302. The fix exposed a downstream case: KCNQ1's record NG\_016178.2 carries only a partial CDS, complement(42918..43038). The old parser read no spans and bio.splice reported 'no CDS spans'; with spans read, the junction map came back 'ok' but empty. We made it fail loud with status 'partial CDS in record'. The splice-region discovery pipeline reads hg38 directly and does not use this parser, so its results are unchanged.

**(F52) Qualifier continuation.** For a qualifier whose value spans lines  $\ell_1, \dots, \ell_m$ ,  $v = \ell_1 \oplus_{\sigma} \ell_2 \oplus_{\sigma} \dots \oplus_{\sigma} \ell_m$  with separator  $\sigma = \emptyset$  for /translation and  $\sigma = \text{one space}$  otherwise; a line starting with / opens a new qualifier only when the running count of quote characters in  $v$  is even.

## 16.13 Core sequence operations

On the same 25 GenBank records, bio.sequence.translate reproduced the /translation qualifier for 285/285 standard-code CDSs (17 CDSs with other genetic codes, codon\_start $\neq$ 1, transl\_except or ribosomal slippage were set aside) and matched Bio.Seq.translate on all 285. Reverse complements agreed on 25/25 whole

records, IUPAC motif search on 192/192 (motif, record) pairs, Wallace  $T_m$  on 500/500 lambda oligos (8–13 nt), and ORF calls ( $\geq 100$  aa, both strands) on 22/22 records (239 = 239 ORFs) against an independent Biopython frame scan. GC fraction agreed to  $10^{-12}$  on 24/25; the exception is the human mitochondrial genome, whose single N sugarcode leaves out of the denominator. Protein masses on 500 UniProt proteins differ from ProteinAnalysis by at most  $5.0e-05$  (relative), from two-decimal residue masses. Reading the code for this comparison turned up a latent bug the data could not: `clean_dna` silently dropped IUPAC ambiguity letters, so `ATGRCTTAA` translated as `ML` (frame shifted) instead of `MX`. No CDS in this set carries an ambiguity code, so the real-data numbers are identical before and after; ambiguity letters now become N (sugarcode 143d1c4), with a regression test. We report it as a code-review finding, not a data finding.

**(F53) Frame preservation.** Cleaning must be a letter-wise map  $c : \Sigma \rightarrow \{A, C, G, T, N\} \cup \{\epsilon\}$  with  $c(x) = \epsilon$  only for non-letters; then codon  $k$  of the cleaned sequence is  $c(s_{3k})c(s_{3k+1})c(s_{3k+2})$  and every downstream codon keeps its frame.

## 16.14 k-mer sets and sketches: exact agreement, a biased estimator

We fetched six coronavirus genomes (SARS-CoV-2, bat RaTG13, pangolin GX-P4L, SARS-CoV Tor2, bat ZC45, MERS) and compared `bio.kmer` with `sourmash` 4.8.14 at  $k = 21$ . With `sourmash` keeping every hash (scaled=1), distinct canonical k-mer counts agreed for 6/6 genomes, and Jaccard and containment agreed exactly on all 15 pairs (maximum difference 0.0). The sketch estimator is a negative result. `bio.kmer`'s minimizer sketch (window  $w = 100$ , about 600 k-mers per genome) under-estimated Jaccard: for SARS-CoV-2 vs RaTG13 the exact value is 0.321, the minimizer estimate 0.236 and `sourmash` `FracMinHash` (scaled=100, about 300 hashes) 0.312. Across the 15 pairs the mean absolute error was 0.0121 for minimizers and 0.0044 for `FracMinHash`, with half the sketch size. Jaccard over minimizer sets is known to be biased because sequence changes shift which k-mer wins each window; `FracMinHash` keeps a k-mer if and only if its hash falls below a fixed threshold, so the kept subset is a uniform sample of the union. The module's docstring already labels the values as estimates; we record the bias and do not recommend minimizer Jaccard for distance estimation.

**(F54) FracMinHash estimator.** With hash  $h$  and threshold  $H/s$ ,  $\text{FMH}_s(A) = \{a \in A : h(a) < H/s\}$  and  $\hat{f} = |\text{FMH}_s(A) \cap \text{FMH}_s(B)| / |\text{FMH}_s(A \cup B)|$ , which is (nearly) unbiased because inclusion depends on the k-mer alone; the Mash distance is  $D = -\frac{1}{k} \ln \frac{2J}{1+J}$ .

## 16.15 Population-genetics statistics on 1000 Genomes

We pulled 1000 Genomes phase 3 genotype counts for 22 well-studied SNPs (APOE, LCT, SLC24A5, HERC2, ALDH2, ABCC11, ADH1B, EDAR, F5, HFE, MTHFR, COMT and others) from the Ensembl GRCh37 REST service, one record per rsID, giving 110 (SNP, super-population) genotype tables. `bio.gstats.hwe_exact` matched an independent full enumeration of the exact HWE distribution on 110/110 tables (maximum relative error  $1.1e-12$ ); 15 tables have  $p < 0.05$ , as expected for a set that includes SNPs with strong population structure inside super-populations. Treating EUR as 'cases' and EAS as 'controls' for 20 SNPs, allelic chi-square (with and without Yates), genotypic chi-square and Woolf odds ratios matched `scipy` and `statsmodels` to  $\leq 1e-13$  relative error, and Benjamini-Hochberg and Bonferroni adjustments matched `statsmodels` exactly (maximum difference  $2e-19$ ). No bug was found in this module.

| SNP        | alleles | EUR (AA/AB/BB) | EAS (AA/AB/BB) | allelic $\chi^2$ | OR       |
|------------|---------|----------------|----------------|------------------|----------|
| rs1426654  | A/G     | 500/3/0        | 0/12/492       | 1954.6           | 2.77e+04 |
| rs16891982 | C/G     | 4/54/445       | 498/6/0        | 1756.6           | 0.000393 |
| rs3827760  | A/G     | 492/11/0       | 12/104/388     | 1516.9           | 622      |
| rs1229984  | T/C     | 1/27/475       | 244/215/45     | 972.8            | 0.0129   |
| rs12913832 | A/G     | 90/186/227     | 502/2/0        | 932.6            | 0.00114  |
| rs17822931 | C/T     | 378/113/12     | 36/150/318     | 840.0            | 22.5     |
| rs4988235  | G/A     | 154/187/162    | 504/0/0        | 686.1            | 0.00048  |
| rs671      | G/N     | 503/0/0        | 349/135/20     | 191.3            | 424      |
| rs53576    | A/G     | 55/243/205     | 212/231/61     | 179.9            | 0.291    |
| rs2231142  | G/T     | 414/83/6       | 251/213/40     | 124.7            | 3.93     |
| rs1799945  | C/G     | 348/137/18     | 478/23/3       | 114.4            | 0.143    |
| rs4680     | G/A     | 133/237/133    | 264/198/42     | 102.7            | 0.388    |
| rs1042713  | G/A     | 183/252/68     | 96/263/145     | 53.7             | 1.94     |
| rs1800562  | G/A     | 461/41/1       | 504/0/0        | 44.0             | 0.011    |
| rs429358   | T/C     | 358/134/11     | 421/79/4       | 22.4             | 0.515    |
| rs1801133  | G/A     | 204/231/68     | 259/192/53     | 10.9             | 0.731    |
| rs7412     | C/T     | 441/61/1       | 407/93/4       | 9.5              | 1.67     |
| rs10427255 | C/T     | 131/236/136    | 94/257/153     | 5.8              | 1.24     |
| rs1815739  | T/C     | 90/257/156     | 98/251/155     | 0.2              | 0.964    |
| rs762551   | C/A     | 58/206/239     | 55/220/229     | 0.1              | 0.967    |

Table 17: EUR vs EAS allele-frequency contrasts from 1000 Genomes phase 3, computed by bio.gstats and matching scipy/statsmodels (benchmarks/sweep\_gstats\_scipy.json).

**(F55) Exact HWE distribution.** For  $n$  individuals with  $n_A$  copies of allele A and  $n_B = 2n - n_A$ , the probability of  $h$  heterozygotes is  $P(h) = \frac{n!}{n_{AA}!h!n_{BB}!} \frac{2^h n_A!n_B!}{(2n)!}$  with  $n_{AA} = (n_A - h)/2$ ,  $n_{BB} = (n_B - h)/2$ ; the two-sided p-value is  $\sum_{h:P(h) \leq P(h_{obs})} P(h)$ .

## 16.16 UniProt lookup: a wrong protein and a cache that never hit

bio.uniprot.search maps a gene symbol to its UniProt entry, and several modules (binding sites, mutation docking) build on it. We checked it against HGNC’s curated symbol-to-UniProt mapping for 60 human genes, fetching each HGNC record separately. Before the fix 59/60 matched. The miss was CDKN2A: the query gene:CDKN2A also matches CDKN2A-AS1, and taking the top hit returned Q9UH64 (the antisense-locus protein) instead of P42771 (p16-INK4a). We switched to gene\_exact and prefer the entry whose primary gene name equals the symbol; after the fix 60/60 match and all 60 returned sequences have the length of the entry fetched by accession. Reading the connector turned up a second bug. The on-disk cache named each file by abs(hash(url)); Python salts string hashes per process, so a new process never found its own cache and offline=True always failed. The same pattern was in the ChEMBL, gnomAD and structure connectors. All four now use SHA-256, and a new process running offline reads TP53 and CDKN2A from cache (P04637, P42771).

**(F56) Stable cache key.** A cache key must be a pure function of the request,  $k(u) = \text{SHA256}(u)$ ; a per-process salted hash  $k_s(u)$  gives  $\Pr[k_{s_1}(u) = k_{s_2}(u)] \approx 2^{-64}$  for independent salts, so the hit rate across processes is effectively zero.

## 16.17 NCBI Gene lookup: Huntington resolved to the serotonin transporter

`bio.entrez.gene_id` resolves a symbol to an NCBI Gene ID and feeds the gene-analysis module. Against HGNC’s curated Gene IDs for the same 60 symbols it scored 58/60. HTT returned 6532, which is SLC6A4 (the serotonin transporter, alias HTT), not huntingtin (3064); TTN returned 7276, which is TTR (transthyretin, alias TTN), not titin (7273). The query field [sym] also matches aliases and the search took the top hit. The function now fetches up to ten hits and returns the one whose official symbol equals the query; after the fix 60/60 match (sugarcode 93d5b12). For a user asking about Huntington disease or titin cardiomyopathy the old answer would have been about the wrong gene.

**(F57) Symbol resolution rule.** Given candidate IDs  $u_1, \dots, u_r$  from an alias-aware search, return  $u^* = u_{\min\{i: \text{sym}(u_i)=q\}}$  if that set is non-empty, else  $u_1$ ; an alias match is used only when no official-symbol match exists.

## 16.18 Reactome pathway mapping

`bio.reactome.pathways_for_uniprot` asks Reactome’s live ContentService which lowest-level pathways contain a protein. As an independent check we used Reactome’s own bulk file UniProt2Reactome.txt for the 61 HGNC-curated UniProt accessions of our 60 genes. A first comparison against canonical-accession rows matched only 59/61: KRAS (P01116) has 9 canonical rows but the service returned 71 pathways, and LMNA (P02545) 2 vs 8. The service pools the isoform rows (P01116-1, P01116-2); pooling them in the file gives 61/61 identical sets (775 = 775 protein-pathway pairs). No bug, but a caveat for users: KRAS4A and KRAS4B pathways come back together.

## 16.19 ChEMBL activities: “best first” from an arbitrary page

`bio.chembl.activities_for_target` promises measured IC50/Ki/Kd values ‘best first’. It fetched the first 50 rows in ChEMBL’s default order and sorted those, so ‘best’ meant best of an arbitrary page. On 12 cancer and CNS targets (between 1,731 and 21,238 activities each) the old top-10 shared at most one molecule with the true top-10, and the reported best was far from the real one; for ABL1 it was 1,200 nM when ChEMBL holds exact measurements at 0.015 nM. The function now asks ChEMBL to sort all activities that carry a pChEMBL value by pChEMBL, descending (sugarcode c6db085). The best value now equals the best pChEMBL-valid measurement on 12/12 targets; this check is partly circular because the reference uses the same ordering, so the evidence for the bug is the before-fix column. Restricting to pChEMBL also drops data-entry artefacts that sort first by raw value: 0.0 nM (JAK2, IDH1),  $5 \times 10^{-9}$  nM (EGFR) and  $-0.63$  nM (DRD2).

**(F58) pChEMBL.**  $\text{pChEMBL} = -\log_{10}(A \cdot 10^{-9})$  for an activity  $A$  in nM with an exact relation; ranking by pChEMBL descending equals ranking by  $A$  ascending on valid records, and  $A \leq 0$  has no pChEMBL.

## 16.20 gnomAD constraint: a version-2 cutoff on version-4 data

`bio.gnomad.gene_constraint` reads pLI and LOEUF from gnomAD r4 and flags a gene as strongly loss-of-function constrained when  $\text{LOEUF} < 0.35$ . That cutoff comes from gnomAD v2. gnomAD’s own help page says v4 o/e and LOEUF values run higher than v2, that v2 thresholds do not transfer, and suggests  $\text{LOEUF} < 0.45$  (the most constrained 15% of 17,063 MANE Select transcripts) for Mendelian interpretation. Under

| target | activities | best before (nM) | best after (nM) | pChEMBL |
|--------|------------|------------------|-----------------|---------|
| EGFR   | 21,238     | 0.45             | 0.01            | 11.00   |
| ABL1   | 6,290      | 1200             | 0.015           | 10.82   |
| BRAF   | 9,702      | 0.3              | 0.02            | 10.70   |
| JAK2   | 18,707     | 1                | 0.0106          | 10.97   |
| FLT3   | 6,684      | 0.8              | 0.01            | 11.00   |
| ERBB2  | 3,534      | 5                | 0.04            | 10.40   |
| PIK3CA | 9,583      | 1                | 0.016           | 10.80   |
| IDH1   | 9,132      | 70               | 0.04            | 10.40   |
| ALK    | 4,300      | 0.4              | 0.07            | 10.15   |
| KIT    | 6,127      | 4                | 0.02            | 10.70   |
| CDK4   | 1,731      | 10               | 0.026           | 10.59   |
| DRD2   | 13,215     | 0.031            | 0.0149          | 10.83   |

Table 18: Best reported activity per target before and after the fix (benchmarks/sweep\_chembl\_order.json).

the old cutoff TP53 (LOEUF 0.418, pLI > 0.999) was called ‘not strongly constrained’. On our 60 genes the old rule flagged 13 and the v4 rule flags 19; the newly flagged genes are APC, FMR1, LMNA, NF1, SCN5A, TP53. Agreement with the independent pLI > 0.9 call rose from 51/60 to 57/60 (sugarcode 0a265e7). The sweep also hit gnomAD’s API rate limit (HTTP 400, ‘try again in a few minutes’); the connector used to retry three times within about 1.5 s, which only burns quota, and now fails fast with a distinct rate-limit error so callers can wait or fall back to the cache.

**(F59) LOEUF.** With observed pLoF count  $O$  and neutral expectation  $E$ ,  $\text{LOEUF} = q_{0.95}(\lambda \mid O)/E$  where the posterior of the true count is  $\lambda \mid O \sim \text{Gamma}(O + 1, 1)$  under a flat prior and Poisson likelihood; a gene is flagged when  $\text{LOEUF} < \tau_v$  with version-specific  $\tau_{v4} = 0.45$ .

## 16.21 Splice-junction maps against ClinVar canonical-site variants

bio.splice.junction\_map builds donor and acceptor windows, keyed by cDNA position, from each gene’s RefSeqGene record, taking the CDS with the most exons as canonical. ClinVar names canonical splice-site variants on the MANE Select transcript (c.N+1, c.N+2, c.N-1, c.N-2), which gives an independent test of both the numbering and the sequence: the named position must be a junction in the map, and the base at the named intronic offset must equal ClinVar’s reference allele. We queried 12 genes; CDH1 is not in the module’s vendored RefSeqGene list and gets an explicit ‘no RefSeqGene junction map’ status rather than a guess (a coverage limit, noted for follow-up). Across the other 11 (TP53, BRCA1, BRCA2, MLH1, MSH2, APC, PTEN, ATM, PALB2, CFTR, NF1) ClinVar returned 161 such SNVs. The map had the junction for 159 and the right reference base for 159; the two misses are named on other transcripts (KCNK12, a neighbouring gene returned by the MSH2 search, and a non-MANE APC isoform). The ‘most exons’ rule picked the MANE transcript for every gene here, NF1 included. A first run of our own checker scored 96 because it assumed 15 intronic bases in the acceptor window; the windows hold 14 intronic bases plus the first exonic base, consistently across all three map builders. No module bug.

**(F60) Window offsets.** For a donor at exon end  $b$  the window is  $x_{b-2} \dots x_{b+6}$ , so intronic position  $+k$  is index  $2 + k$ ; for an acceptor at exon start  $c$  the window is  $x_{c-14} \dots x_c$ , so intronic position  $-k$  is index  $14 - k$  (0-based); on the minus strand both windows are reverse-complemented before indexing.

## 16.22 CFD off-target scores

modules.cfd\_offtarget scores guide/off-target pairs with the Cutting Frequency Determination model. We re-implemented the scoring loop of the authors' published calculator on their released matrices (mismatch\_score.pkl, pam\_scores.pkl) and scored the 600 BRCA1 guide/off-target pairs against all 16 PAM dinucleotides: 9,600/9,600 scores agree, maximum difference 2.2e-16. No bug.

**(F61) CFD score.**  $\text{CFD}(g, o, p) = \pi(p_2 p_3) \prod_{i: g_i \neq o_i} m(\text{rg}_i: d\overline{o}_i, i)$ , where  $\overline{o}_i$  is the complement of the off-target base,  $m$  the position-specific mismatch activity and  $\pi$  the PAM activity.

## 16.23 Short tandem repeats: a flank homopolymer hid compound repeats

modules.str\_scope.find\_strs finds tandem repeats with 1–6 bp units. We ran it and pytrf 1.5.0 (the finder behind Krait and TRF-style pipelines) with pytrf's default minimum repeat counts (12/7/5/4/4/4 for mono- to hexanucleotides) on all 25 GenBank records in our dataset set, from phiX174 to a 193,689-bp RefSeqGene region. Before the fix sugarcode found 208 of pytrf's 211 repeats and nothing pytrf missed. The three losses had one cause: the scan moved left to right and, after taking a short homopolymer, jumped to the end of that run, past the start of an overlapping longer repeat. In HSV-1 (NC\_001806.2) a seven-base G run swallowed the start of (GGGGTT)<sub>4</sub>; in NG\_005905.2 T and A runs did the same to (TTTTG)<sub>4</sub> and (AAAT)<sub>4</sub>. The finder now takes every maximal periodic run at every unit length and resolves overlaps longest first, trying the other phases of a run before dropping it (sugarcode 727344a). After the fix it finds 211/211, with 0 extra calls; 209 have identical coordinates and the other 2 are the same span reported in the disease-motif phase (CGG rather than GCG in PTEN's 5' UTR), which the module does on purpose. The same module labels repeat genotypes. The old rule used only the distance from a normal upper bound (normal if  $\leq 2$  above it, intermediate up to 10 above, else 'pathogenic-range candidate'). Compared with the GeneReviews allele tables it mislabels HTT CAG counts 27, 28, 36 (27 and 28 are intermediate, 36 is reduced-penetrance pathogenic) and 154 FMR1 CGG counts between 5 and 300, from 47 to 200: every premutation allele (55–200) was called a pathogenic expansion, which in counselling terms confuses a carrier with an affected full-mutation allele. expansion\_call now takes an optional locus and uses the published bands for HTT and FMR1 (0 disagreements on HTT 5–120); without a locus the old behaviour is unchanged.

**(F62) Maximal periodic runs.** For unit length  $u$ , position  $j$  is periodic if  $x_j = x_{j+u}$ ; a maximal run of periodic positions  $[a, b]$  defines the span  $[a, b + u]$  holding  $\lfloor (b + u - a - k)/u \rfloor$  whole units from phase  $k \in \{0, \dots, u - 1\}$ . Candidates over all  $u$  are kept greedily by length (ties: smaller  $u$ , then leftmost) subject to disjoint whole-unit intervals, which is the rule pytrf-style finders apply implicitly by reporting the longest repeat at each locus.

## 16.24 Bayesian ACMG scoring against ClinGen expert panels

modules.acmg\_bayesian scores ACMG/AMP criteria with the Tavtigian et al. (2018) Bayesian model in its point form (supporting 1, moderate 2, strong 4, very strong 8; pathogenic  $\geq 10$ , likely pathogenic 6–9, likely benign  $-1$  to  $-6$ , benign  $\leq -7$ ). The ClinGen Evidence Repository publishes, for every variant a Variant Curation Expert Panel (VCEP) has classified, the criteria it applied and its final call. We used 7,929 current classifications from 33 panels across 117 genes (the server closed the bulk download after about 18 MB, so only complete rows were used). ClinGen writes modified strengths as suffixes: PM2.Supporting, PVS1.Strong, PM3.Very Strong, BS1.Stand Alone. The module accepted a strength only as a separate field,

so in 5,699 rows at least one code was rejected, and it reproduced 4,583/7,929 panel calls. The rejections were listed in the output rather than hidden, but anyone pasting ClinGen notation got a wrong class. The parser now reads the suffix, and a stand-alone benign strength acts like BA1 (sugarcode cdf160b). The module now reproduces 7,515/7,929 calls (94.8%), with no rejected codes. As a baseline we coded the original ACMG/AMP 2015 combining rules (Richards et al., Table 5), counting each criterion at its applied strength and calling conflicting evidence uncertain. They reproduce 7,267/7,929 (91.7%); the point model is closer to what the panels actually do. The largest remaining disagreement is 229 variants the panels call uncertain and the model calls likely benign; 177 of them net to exactly  $-1$  point, typically a lone BP4. The 2015 rules need two benign supporting criteria for likely benign, and the panels follow that; the 2018 posterior bands do not. We report this as a known difference and did not change the bands. BA1 with pathogenic criteria also met is rare and panels treat it differently (21 benign, 3 pathogenic, 3 uncertain significance, 4 likely benign), so the module keeps BA1 as an override but flags the conflict.

**(F63) Point form of the Bayesian model.** With prior  $\pi = 0.1$  and  $O_{PVSt} = 350$ , a criterion of strength  $k$  points has odds  $350^{k/8}$ ; summed points  $S$  give posterior  $P(S) = \frac{\frac{\pi}{1-\pi} 350^{S/8}}{1 + \frac{\pi}{1-\pi} 350^{S/8}}$ , so  $P(10) = 0.994$ ,  $P(6) = 0.900$ ,  $P(-1) = 0.051$  and  $P(-7) = 0.0007$ .

## 16.25 ClinVar interpretation: exon-spanning ranges and review stars

modules.openclinvar infers a variant’s molecular consequence from its HGVS name when the name alone decides it, and abstains otherwise. We pulled 720 ClinVar records (60 per gene for BRCA1, BRCA2, TP53, MLH1, MSH2, CFTR, APC, PTEN, LDLR, MYH7, SCN5A and RYR1) and scored the module’s call as correct when ClinVar lists that consequence for the record. Of 682 records with a ClinVar consequence the module got 646 right, abstained on 32 and was wrong on 4. All four were ranges judged by their first coordinate only: c.453+625\_545+920delinsT in MLH1 starts in an intron but removes the exon between two introns, c.5468-64\_5480dup in BRCA1 runs from the intron into the exon, and c.-19\_\*21del in TP53 deletes the whole coding sequence; the module called them intronic or UTR, the lowest-weight classes in its evidence model. Ranges are now read at both ends: ends in different exon/intron segments mean a boundary is crossed (splice disruption) and a range from the 5’ UTR into the CDS removes the start codon (sugarcode 7067655). After the fix: 651 right, 30 abstentions, 1 wrong; the remaining one (MSH2 c.2459-2\_2472del) is annotated by ClinVar on an older transcript, and c.2459-2 is an acceptor-site base, so we kept the call. clinvar\_stars maps a review status to ClinVar’s 0–4 gold stars, which weight ClinVar evidence in the module’s trace. Against the 9 statuses in ClinVar’s review-status table it was wrong once: ‘criteria provided, multiple submitters’ (the status used for somatic clinical-impact and oncogenicity classifications) got one star instead of two, because the rule required the germline suffix ‘no conflicts’. Now 9/9.

**(F64) Segment rule for c. ranges.** Write each end as  $(a, s, k)$ : anchor  $a$  (with region 5’ UTR, CDS or 3’ UTR), intron side  $s \in \{+, -, \emptyset\}$  and offset  $k$ . Two ends lie in the same segment iff both are exonic, or both are intronic with  $(a_1, s_1) = (a_2, s_2)$  or  $s_1 = +, s_2 = -, a_2 = a_1 + 1$ . A range whose ends lie in different segments crosses at least one splice site.

## 16.26 CRISPR on-target scores against measured guide activity

modules.crispr\_opt has four on-target scorers: Rule Set 2 (the Azimuth gradient-boosted model, with or without the peptide-position features), Rule Set 1 (Doench 2014 logistic weights), and a hand-built heuristic

used only when a guide sits too close to the sequence edge for a 30-mer context. We scored all 5,310 guides in the Doench 2016 FC+RES set (17 genes; file from the Azimuth repository) and compared with the measured, within-gene rank-normalised activity. Spearman correlations: Rule Set 2 with position 0.7106, sequence-only 0.6693, Rule Set 1 0.3948, heuristic 0.1125; the repository’s own prediction column gives 0.7148. Median per-gene values are 0.6965, 0.6599, 0.468 and 0.1357. Both published models were trained on these guides, so their numbers are in-sample and flattering; the heuristic was not fitted and is close to useless ( $\rho \approx 0.11$ ). Two consequences for users. `design_guides` ranks with Rule Set 1 even though Rule Set 2 ships in the same module; we have not switched it, because the case for switching needs a held-out screen and we have not run one. And the edge fallback should be read as ‘no reliable score’, which the module already labels in its `on_target_model` field (`heuristic_edge_fallback`). No code bug was found.

**(F65) Spearman correlation.** For scores  $s$  and measurements  $y$  with ranks  $R(s_i), R(y_i)$ ,  $\rho = \text{corr}(R(s), R(y))$ , which for untied data equals  $1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)}$  with  $d_i = R(s_i) - R(y_i)$ ; the per-gene median guards against between-gene differences in dynamic range.

## 16.27 Replicate planning: a two-entry z table

`modules.gene_analysis.power_estimate` tells a user how many replicates per group an experiment needs. It added 0.84 to 1.96 when the target power was at most 0.8 and 1.28 otherwise, which is right only at 0.8 and 0.9. Against statsmodels’ NormalIndPower solver on 41 combinations of power (0.5–0.99), effect and variance, the old rule matched 7 times and over-asked by up to  $2.03\times$  (power 0.5 needs 8 replicates for a one-SD effect, it said 16); at 0.95 and 0.99 it under-asked. The function now uses exact normal quantiles and takes alpha as a parameter (sugarcode 4defcdb); it matches statsmodels in 41/41 cases. This is a formula check, not a real-data sweep, and the rest of `gene_analysis` is not yet validated.

**(F66) Two-sample size.** For a two-sided test at level  $\alpha$  with power  $1 - \beta$ , difference  $\delta$  and per-observation variance  $\sigma^2$ ,  $n = \lceil 2\sigma^2(z_{1-\alpha/2} + z_{1-\beta})^2 / \delta^2 \rceil$  per group.

## 17 Adding evolutionary conservation to the splice-region model

The best splice-region model so far is a logistic regression on PWM and MaxEntScan features. Neither feature sees conservation. We fetched phyloP100way at each variant position from the UCSC Genome Browser REST API for a stratified subsample of 1,501 of the 9,235 modelled variants (379 pathogenic, 1,054 genes, 0 missing). The subsample keeps the class-by-site proportions of the full set; the API rate limit (one request per second) is why we did not fetch all 9,235. Median phyloP is 3.214 at pathogenic and 0.191 at benign positions. On its own, phyloP reaches AUROC 0.8377 (0.9108 at donors, 0.6966 at acceptors). Added to logit+MaxEnt under gene-grouped 5-fold cross-validation repeated over 20 seeds, AUROC rises from 0.963 to 0.9678. A paired bootstrap on the averaged out-of-fold predictions gives a 95% interval for the gain of  $[+0.0008, +0.0091]$ , which excludes zero. The gain is concentrated at donor-region positions (+3 to +6), where the intronic consensus is short and conservation carries information the PWM misses. At acceptors (the polypyrimidine tract, −3 to −14), MaxEntScan already scores well and conservation adds little.

| site     | $n$  | phyloP alone | logit+MaxEnt | + phyloP |
|----------|------|--------------|--------------|----------|
| donor    | 876  | 0.9108       | 0.9557       | 0.9634   |
| acceptor | 625  | 0.6966       | 0.9702       | 0.9723   |
| all      | 1501 | 0.8377       | 0.9630       | 0.9678   |

Table 19: AUROC with and without phyloP (discovery/splice\_region\_vus/phylop\_lift.json). Per-site values use the seed-averaged predictions.

**Caveat.** ClinVar submitters may cite conservation as supporting evidence (ACMG criteria PP3/BP4). Some of the gain may therefore be label circularity rather than new biology. We report the gain as modest and conditional on that caveat.

**(F47) Model with conservation.**  $\text{logit} P(\text{path} \mid x) = \beta_0 + \beta^\top \mathbf{z}(x) + \gamma \phi(x) + \delta \phi(x) \mathbb{I}[\text{donor}]$ , with  $\mathbf{z}$  the standardised PWM/MaxEnt/position features and  $\phi$  the phyloP score; the interaction lets conservation matter differently at donors.

**(F48) Paired bootstrap.** For resamples  $b = 1..B$  of variant indices  $I_b$ ,  $\Delta_b = \text{AUC}(y_{I_b}, \hat{p}_{I_b}^+) - \text{AUC}(y_{I_b}, \hat{p}_{I_b}^0)$ ; the interval is the 2.5th and 97.5th percentile of  $\{\Delta_b\}$ . Pairing on the same resample removes the shared sampling noise of the two models.

## 18 Discussion

Validating each module against an independent reference exposed three kinds of result: implementations that match their reference exactly (CAI vs Biopython, CYP2D6 translation vs CPIC after the fix), implementations whose defaults were stale or weak (CPIC activity values, genome-wide CAI reference, Nussinov folding), and places where a strong published tool beats our model (SpliceAI on splice-region variants). Each fix is committed with the result file that motivated it. The CNN is useful as a calibrated, cheap second opinion at high thresholds, not as a replacement for SpliceAI.

## 19 Conclusion

The sweep so far corrected two clinically relevant defaults (CPIC activity values, clopidogrel handling of “Likely” phenotypes), improved CAI correlation with measured abundance, and replaced base-pair maximisation with thermodynamic folding. The accession-level dataset target (120) and the formula target (10) are met; the external-tool target (40) is met (manifests/tools.tsv). The manifests in the appendix list exactly what was used.

## 20 Module inventory and validation coverage

sugarcode-ai contains 95 application modules and 32 bio/ libraries. 47 of these 127 have been checked against an independent reference on real data in this work (16 modules, 31 libraries). The rest are not yet validated. That does not mean they are wrong, only that no claim about them is made here. Validation status is a curated list tied to committed result files (manifests/module\_inventory.tsv, built by sweep/module\_inventory.py), not a keyword match.

| kind   | name             | py lines | status    |
|--------|------------------|----------|-----------|
| bio    | align            | 164      | validated |
| bio    | bed              | 180      | validated |
| bio    | chembl           | 184      | validated |
| bio    | codon            | 170      | validated |
| bio    | de               | 218      | validated |
| bio    | entrez           | 196      | validated |
| bio    | fastq            | 183      | validated |
| bio    | genbank          | 103      | validated |
| bio    | gff              | 237      | validated |
| bio    | gnomad           | 122      | validated |
| bio    | gstats           | 194      | validated |
| bio    | kmer             | 261      | validated |
| bio    | motif            | 198      | validated |
| bio    | newick           | 248      | validated |
| bio    | orf              | 189      | validated |
| bio    | pdb              | 346      | validated |
| bio    | phylo            | 232      | validated |
| bio    | pileup           | 148      | validated |
| bio    | primer           | 278      | validated |
| bio    | proteinprops     | 261      | validated |
| bio    | pwm              | 63       | validated |
| bio    | reactome         | 63       | validated |
| bio    | restriction      | 150      | validated |
| bio    | rnaseq           | 201      | validated |
| bio    | sam              | 221      | validated |
| bio    | sequence         | 174      | validated |
| bio    | splice           | 682      | validated |
| bio    | stockholm        | 260      | validated |
| bio    | structures       | 267      | validated |
| bio    | uniprot          | 106      | validated |
| bio    | vcf              | 263      | validated |
| module | acmg_bayesian    | 469      | validated |
| module | cf_d_offtarget   | 157      | validated |
| module | chem_descriptors | 923      | validated |
| module | chem_similarity  | 234      | validated |
| module | codon_opt        | 289      | validated |
| module | crispr_opt       | 648      | validated |
| module | crisprscan_score | 178      | validated |
| module | deepsplice       | 751      | validated |
| module | docking_studio   | 535      | validated |
| module | mit_offtarget    | 154      | validated |
| module | neohunter        | 358      | validated |
| module | openclinvar      | 514      | validated |
| module | pgx_guidelines   | 287      | validated |
| module | profile_hmm      | 747      | validated |
| module | rna_nussinov     | 387      | validated |
| module | str_scope        | 340      | validated |
| bio    | fasta            | 55       | not yet   |
| module | alpha_fold_ui    | 404      | not yet   |
| module | bio_copilot      | 245      | not yet   |
| module | bio_material     | 114      | not yet   |
| module | bio_switch       | 511      | not yet   |
| module | biofactory_1_a   | 102      | not yet   |
| module | biogpt_lit       | 249      | not yet   |
| module | bioimage_ai      | 146      | not yet   |
| module | bioplayground    | 98       | not yet   |
| module | bioprint_pro     | 143      | not yet   |
| module | biosimvr         | 200      | not yet   |
| module | car_t_designer   | 104      | not yet   |
| module | cell_free_opt    | 107      | not yet   |
| module | cell_twin        | 170      | not yet   |
| module | cellfatenet      | 199      | not yet   |
| module | cellpainter      | 289      | not yet   |
| module | cellpainter_4d   | 164      | not yet   |
| module | chemgpt_engine   | 305      | not yet   |
| module | crispr_cargo     | 156      | not yet   |
| module | crispr_muse      | 278      | not yet   |

| kind   | name                  | py lines | status  |
|--------|-----------------------|----------|---------|
| module | crisprater            | 97       | not yet |
| module | dark_genome           | 202      | not yet |
| module | dna_to_code           | 111      | not yet |
| module | dti_bench             | 196      | not yet |
| module | ecosystem             | 175      | not yet |
| module | enterprise_bio        | 260      | not yet |
| module | epi_edit              | 178      | not yet |
| module | evidence_mining       | 418      | not yet |
| module | evofold_4d            | 236      | not yet |
| module | fate_predictor        | 131      | not yet |
| module | gene_analysis         | 262      | not yet |
| module | gene_explorer         | 140      | not yet |
| module | gene_tx_opt           | 149      | not yet |
| module | genomegpt             | 279      | not yet |
| module | infinite_diagnosis    | 130      | not yet |
| module | liquid_biopsy         | 376      | not yet |
| module | living_computer       | 857      | not yet |
| module | living_tx             | 184      | not yet |
| module | metabodesigner        | 170      | not yet |
| module | micro_tx              | 218      | not yet |
| module | microaiverse          | 138      | not yet |
| module | microbiome_exp        | 119      | not yet |
| module | microbiome_rx         | 172      | not yet |
| module | molecule_eval         | 149      | not yet |
| module | mutdock               | 337      | not yet |
| module | neodti_engine         | 257      | not yet |
| module | neuro_hub             | 44       | not yet |
| module | neuro_hub_dashboard   | 25       | not yet |
| module | neuro_pipeline        | 326      | not yet |
| module | neuroplan_ai          | 449      | not yet |
| module | nexus_support         | 299      | not yet |
| module | omega_stats           | 161      | not yet |
| module | oncocircuit           | 124      | not yet |
| module | organoid_ai           | 216      | not yet |
| module | organoid_screen       | 275      | not yet |
| module | pdx_insight           | 138      | not yet |
| module | phage_designer        | 75       | not yet |
| module | phage_tx              | 161      | not yet |
| module | phageforge            | 87       | not yet |
| module | prime_design          | 248      | not yet |
| module | promoter_lib          | 143      | not yet |
| module | proteinPainter        | 161      | not yet |
| module | qsar_bench            | 353      | not yet |
| module | rarenet_ai            | 365      | not yet |
| module | report_studio         | 213      | not yet |
| module | riboswitch            | 142      | not yet |
| module | rna_decoder           | 120      | not yet |
| module | robotic_flow          | 128      | not yet |
| module | stability_ai          | 199      | not yet |
| module | structural_biophysics | 512      | not yet |
| module | syn_bio_studio        | 43       | not yet |
| module | syn_stab_ai           | 69       | not yet |
| module | synbio_studio         | 113      | not yet |
| module | synbio_wizard         | 224      | not yet |
| module | syndroid              | 231      | not yet |
| module | synlife_evo           | 278      | not yet |
| module | synthetic_life        | 197      | not yet |
| module | tissue_eng            | 193      | not yet |
| module | vector_opt            | 95       | not yet |
| module | virtual_cell          | 504      | not yet |

## **A Tools Table**

| Tool                                 | Version                | Status/used in                                                                             |
|--------------------------------------|------------------------|--------------------------------------------------------------------------------------------|
| pyhmmer                              | 0.12.3                 | installed                                                                                  |
| ViennaRNA                            | 2.7.2                  | installed                                                                                  |
| RDKit                                | 2026.3.6               | installed                                                                                  |
| SpliceAI                             | 1.3.1                  | installed                                                                                  |
| TensorFlow                           | 2.15.1                 | installed                                                                                  |
| Keras                                | 2.15.0                 | installed                                                                                  |
| scikit-learn                         | 1.7.2                  | installed                                                                                  |
| SciPy                                | 1.15.3                 | installed                                                                                  |
| NumPy                                | 1.26.4                 | installed                                                                                  |
| Biopython                            | 1.88                   | installed                                                                                  |
| twobitreader                         | 4.0.2                  | installed                                                                                  |
| AutoDock Vina (python)               | 1.2.7                  | installed                                                                                  |
| Open Babel                           | 3.1.1.23               | installed                                                                                  |
| pytest                               | 9.1.1                  | installed                                                                                  |
| pandas                               | 2.3.3                  | installed                                                                                  |
| MHCflurry                            | 2.2.1                  | sweep/neohunter_mhcflurry_cmp.py                                                           |
| CRISPOR reference scorers (vendored) | web service/vendored   | sweep/_crispor_*_ref.py                                                                    |
| EBI Clustal Omega REST               | web service/vendored   | data/README.md                                                                             |
| CPIC API                             | web service/vendored   | sweep/pgx_cpic_bench.py                                                                    |
| NCBI E-utilities                     | web service/vendored   | src/sc_validate                                                                            |
| LOVD3 API                            | web service/vendored   | discovery/splice_region_vus/lovd_top20.js                                                  |
| MaxEntScan (maxentpy)                | 0.0.4                  | discovery/splice_region_vus/maxent_cmp.py                                                  |
| codon-bias                           | 0.5.0                  | sweep/codon_tai_enc.py                                                                     |
| statsmodels                          | 0.15.0                 | benchmarks/sweep_codon_cai_multispecies (BH-FDR)                                           |
| PyTorch                              | 2.14.0+cpu             | discovery/splice_region_vus/cnn_cv.py, cnn (splice CNN)                                    |
| Matplotlib                           | 3.10.9                 | paper/make_figs.py (all figures)                                                           |
| FreeSASA                             | 2.x (python)           | sweep/sasa_vs_freesasa.py -i<br>marks/sweep_sasa_freesasa.json                             |
| primer3-py                           | 2.3.1                  | sweep/primer_vs_primer3.py -i<br>marks/sweep_primer_primer3.json;<br>bio.primer.hairpin_dg |
| DendroPy                             | 5.1.0                  | sweep/phylo_vs_dendropy.py -i<br>marks/sweep_phylo_dendropy.json                           |
| orfipy                               | 0.0.4                  | sweep/orf_vs_orfipy.py -i<br>marks/sweep_orf_orfipy.json                                   |
| pysam (htslib)                       | 0.24.1                 | sweep/vcf_vs_pysam.py -i<br>marks/sweep_vcf_pysam.json (remote t<br>of ClinVar VCF)        |
| PyDESeq2                             | 0.5.2                  | sweep/de_vs_pydeseq2.py -i<br>marks/sweep_de_pydeseq2.json;<br>bio.de.de_analysis_deseq2   |
| parasail                             | 2.6.1                  | sweep/align_vs_parasail.py -i<br>marks/sweep_align_parasail.json                           |
| g:Profiler g:GOSl API                | e114_eg62_p19_37140483 | sweep/dex_enrichment_gprofiler.py -i<br>marks/sweep_dex_enrichment.json                    |
| STRING API                           | 12.0                   | sweep/dex_string_ppi.py -i<br>marks/sweep_dex_string_ppi.json                              |
| Reactome AnalysisService             | live 2026-09-24        | sweep/dex_reactome.py -i<br>marks/sweep_dex_reactome.json                                  |
| networkx                             | local 3.4.2            | sweep/dex_string_network.py -i<br>marks/sweep_dex_string_network.json                      |
| gseapy                               | local 1.3.1            | sweep/dex_gsea_hallmark.py -i                                                              |

## B Dataset Manifest

| #  | Dataset                                                                        | Accession/version                                               | Used in                                                   |
|----|--------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------|
| 1  | ClinVar variant_summary                                                        | variant_summary.txt.gz (GRCh38)                                 | data/clinvar_sample_500.tsv; discovery/splice_region_vus/ |
| 2  | UCSC hg38 reference genome                                                     | hg38.2bit                                                       | discovery/splice_region_vus/extract.py                    |
| 3  | LOVD3 shared variant records                                                   | top-20 VUS query                                                | discovery/splice_region_vus/lovd_top20.json               |
| 4  | SKEMPI 2.0                                                                     | skempi_v2.csv (PMID 30598530)                                   | data/skempi_v2.csv                                        |
| 5  | BiGG e.coli_core                                                               | e.coli_core (PMID 21988831)                                     | data/e.coli_core.json                                     |
| 6  | NCBI RefSeqGene BRCA1                                                          | NG_005905.2:60000-70000                                         | data/brca1_10kb.fa; data/crispr_fresh_pairs.tsv           |
| 7  | CRISPOR reference CFD/MIT matrices                                             | crisporWebsite (PMID 30341956)                                  | data/crispr_cfd_oracle.tsv; sweep/_crispor_mit_ref.py     |
| 8  | CRISPOR Rule Set 1 reference                                                   | crisporEffScores.py                                             | data/doench_rs1_oracle.tsv                                |
| 9  | UniProtKB reviewed globins                                                     | 25 sequences                                                    | data/globins25.fa                                         |
| 10 | UniProtKB reviewed non-globins                                                 | 25 sequences                                                    | data/nonglobins25.fa                                      |
| 11 | Pfam Globin HMM                                                                | PF00042.29                                                      | data/pfam_PF00042_globin.hmm                              |
| 12 | Rfam tRNA seed                                                                 | RF00005                                                         | data/rfam/RF00005.sto                                     |
| 13 | Rfam 5S rRNA seed                                                              | RF00001                                                         | data/rfam/RF00001.sto                                     |
| 14 | Rfam RNase P RNA seed                                                          | RF00010                                                         | data/rfam/RF00010.sto                                     |
| 15 | ChEMBL approved drugs                                                          | 1000-compound slice                                             | data/chembl_approved_1000.tsv                             |
| 16 | Kazusa CUTG E. coli K12                                                        | taxid 83333                                                     | data/README.md (sanity only)                              |
| 17 | NCBI RefSeq E. coli K-12 MG1655 CDS                                            | GCF_000005845.2                                                 | sweep/codon_cai_bench.py                                  |
| 18 | PaxDb E. coli integrated abundance                                             | 511145-WHOLE_ORGANISM-integrated                                | sweep/codon_cai_bench.py                                  |
| 19 | CPIC CYP2C19 diplotype table                                                   | 666 diplotypes                                                  | sweep/pgx_cpic_bench.py                                   |
| 20 | CPIC CYP2D6 diplotype table                                                    | 16836 diplotypes                                                | sweep/pgx_cpic_bench.py                                   |
| 21 | CPIC CYP2D6 allele activity table                                              | 208 alleles                                                     | sugarcode-ai pgx_guidelines/data                          |
| 22 | Edinburgh Genome Foundry codon tables                                          | e.coli_316407                                                   | sweep/codon_cai_bench.py (via sugarcode-ai)               |
| 23 | IEDB MHC-I binding data 2013 (Kim et al. 2014)                                 | bdata.20130222.mhci.txt                                         | sweep/neohunter_iedb_bench.py                             |
| 24 | LP-PDBBind (Li et al. leak-proof split)                                        | LP.PDBBind.csv                                                  | sweep/docking_bench.py                                    |
| 25 | NCBI RefSeq transcripts (TP53, BRCA1, BRCA2, BRAF, EGFR)                       | NM_000546.6, NM_007294.4, NM_000059.4, NM_004333.6, NM_005228.5 | sweep/primer_vs_primer3.py                                |
| 26 | ClinVar GRCh38 VCF (tabix regions TP53, BRCA1, BRCA2, BRAF, EGFR)              | clinvar.vcf.gz (46,603 records)                                 | sweep/vcf_vs_pysam.py                                     |
| 27 | GEO GSE52778 airway smooth muscle RNA-seq (Himes et al. 2014), NCBI raw counts | GSE52778_raw_counts_GRCh38.p13NCBI raw counts                   | sweep/ncbi_raw_counts.py                                  |

## C Accession-Level Dataset Manifest

Each row is one identifier-backed record fetched individually and used in a committed analysis (counting rule in manifests/README.md).

| # | Kind  | Study | Accession                       | Used in                                                   |
|---|-------|-------|---------------------------------|-----------------------------------------------------------|
| 1 | study | 1     | variant_summary.txt.gz (GRCh38) | data/clinvar_sample_500.tsv; discovery/splice_region_vus/ |
| 2 | study | 2     | hg38.2bit                       | discovery/splice_region_vus/extract.py                    |
| 3 | study | 3     | top-20 VUS query                | discovery/splice_region_vus/lovd_top20.json               |
| 4 | study | 4     | skempi_v2.csv (PMID 30598530)   | data/skempi_v2.csv                                        |
| 5 | study | 5     | e.coli_core (PMID 21988831)     | data/e.coli_core.json                                     |
| 6 | study | 6     | NG_005905.2:60000-70000         | data/brca1_10kb.fa; data/crispr_fresh_pairs.tsv           |
| 7 | study | 7     | crisporWebsite (PMID 30341956)  | data/crispr_cfd_oracle.tsv; sweep/_crispor_mit_ref.py     |

| #  | Kind      | Study | Accession                                | Used in                                        |
|----|-----------|-------|------------------------------------------|------------------------------------------------|
| 8  | study     | 8     | crisporEffScores.py                      | data/doench_rsl_oracle.tsv                     |
| 9  | study     | 9     | 25 sequences                             | data/globins25.fa                              |
| 10 | study     | 10    | 25 sequences                             | data/nonglobins25.fa                           |
| 11 | study     | 11    | PF00042.29                               | data/pfam.PF00042_globin.hmm                   |
| 12 | study     | 12    | RF00005                                  | data/rfam/RF00005.sto                          |
| 13 | study     | 13    | RF00001                                  | data/rfam/RF00001.sto                          |
| 14 | study     | 14    | RF00010                                  | data/rfam/RF00010.sto                          |
| 15 | study     | 15    | 1000-compound slice                      | data/chembl-approved_1000.tsv                  |
| 16 | study     | 16    | taxid 83333                              | data/README.md (sanity only)                   |
| 17 | study     | 17    | GCF_000005845.2                          | sweep/codon_cai_bench.py                       |
| 18 | study     | 18    | 511145-<br>WHOLE_ORGANISM-<br>integrated | sweep/codon_cai_bench.py                       |
| 19 | study     | 19    | 666 diplotypes                           | sweep/pgx_cplic_bench.py                       |
| 20 | study     | 20    | 16836 diplotypes                         | sweep/pgx_cplic_bench.py                       |
| 21 | study     | 21    | 208 alleles                              | sugarcode-ai pgx_guidelines/data               |
| 22 | study     | 22    | e.coli_316407                            | sweep/codon_cai_bench.py (via<br>sugarcode-ai) |
| 23 | study     | 23    | bdata.20130222.mhci.txt                  | sweep/neohunter_iedb_bench.py                  |
| 24 | study     | 24    | LP_PDBBind.csv                           | sweep/docking_bench.py                         |
| 25 | pdb_entry |       | 1CNW                                     | sweep/docking_bench.py                         |
| 26 | pdb_entry |       | 1EJN                                     | sweep/docking_bench.py                         |
| 27 | pdb_entry |       | 1F5L                                     | sweep/docking_bench.py                         |
| 28 | pdb_entry |       | 1G54                                     | sweep/docking_bench.py                         |
| 29 | pdb_entry |       | 1GI1                                     | sweep/docking_bench.py                         |
| 30 | pdb_entry |       | 1K1J                                     | sweep/docking_bench.py                         |
| 31 | pdb_entry |       | 1O3L                                     | sweep/docking_bench.py                         |
| 32 | pdb_entry |       | 1P1O                                     | sweep/docking_bench.py                         |
| 33 | pdb_entry |       | 1Y3X                                     | sweep/docking_bench.py                         |
| 34 | pdb_entry |       | 1ZGI                                     | sweep/docking_bench.py                         |
| 35 | pdb_entry |       | 2ERZ                                     | sweep/docking_bench.py                         |
| 36 | pdb_entry |       | 2J4I                                     | sweep/docking_bench.py                         |
| 37 | pdb_entry |       | 2P4J                                     | sweep/docking_bench.py                         |
| 38 | pdb_entry |       | 2X7T                                     | sweep/docking_bench.py                         |
| 39 | pdb_entry |       | 2XAB                                     | sweep/docking_bench.py                         |
| 40 | pdb_entry |       | 2XJG                                     | sweep/docking_bench.py                         |
| 41 | pdb_entry |       | 2ZDN                                     | sweep/docking_bench.py                         |
| 42 | pdb_entry |       | 3B3C                                     | sweep/docking_bench.py                         |
| 43 | pdb_entry |       | 3C84                                     | sweep/docking_bench.py                         |
| 44 | pdb_entry |       | 3D8W                                     | sweep/docking_bench.py                         |
| 45 | pdb_entry |       | 3HKQ                                     | sweep/docking_bench.py                         |
| 46 | pdb_entry |       | 3HLL                                     | sweep/docking_bench.py                         |
| 47 | pdb_entry |       | 3IBI                                     | sweep/docking_bench.py                         |
| 48 | pdb_entry |       | 3M35                                     | sweep/docking_bench.py                         |
| 49 | pdb_entry |       | 3PO6                                     | sweep/docking_bench.py                         |
| 50 | pdb_entry |       | 3R17                                     | sweep/docking_bench.py                         |
| 51 | pdb_entry |       | 3R4N                                     | sweep/docking_bench.py                         |
| 52 | pdb_entry |       | 4B2I                                     | sweep/docking_bench.py                         |
| 53 | pdb_entry |       | 4DB7                                     | sweep/docking_bench.py                         |
| 54 | pdb_entry |       | 4F5Y                                     | sweep/docking_bench.py                         |
| 55 | pdb_entry |       | 4FL1                                     | sweep/docking_bench.py                         |
| 56 | pdb_entry |       | 4J93                                     | sweep/docking_bench.py                         |
| 57 | pdb_entry |       | 4JFM                                     | sweep/docking_bench.py                         |
| 58 | pdb_entry |       | 4KNJ                                     | sweep/docking_bench.py                         |
| 59 | pdb_entry |       | 4LXZ                                     | sweep/docking_bench.py                         |
| 60 | pdb_entry |       | 4MO8                                     | sweep/docking_bench.py                         |
| 61 | pdb_entry |       | 4NH7                                     | sweep/docking_bench.py                         |
| 62 | pdb_entry |       | 4O04                                     | sweep/docking_bench.py                         |
| 63 | pdb_entry |       | 4OAG                                     | sweep/docking_bench.py                         |
| 64 | pdb_entry |       | 4R76                                     | sweep/docking_bench.py                         |
| 65 | pdb_entry |       | 4RFC                                     | sweep/docking_bench.py                         |
| 66 | pdb_entry |       | 4TIM                                     | sweep/docking_bench.py                         |
| 67 | pdb_entry |       | 4XIT                                     | sweep/docking_bench.py                         |
| 68 | pdb_entry |       | 4YTC                                     | sweep/docking_bench.py                         |
| 69 | pdb_entry |       | 4YXI                                     | sweep/docking_bench.py                         |
| 70 | pdb_entry |       | 4ZME                                     | sweep/docking_bench.py                         |
| 71 | pdb_entry |       | 5AML                                     | sweep/docking_bench.py                         |

| #   | Kind      | Study | Accession                              | Used in                         |
|-----|-----------|-------|----------------------------------------|---------------------------------|
| 72  | pdb_entry |       | 5CAP                                   | sweep/docking_bench.py          |
| 73  | pdb_entry |       | 5CAU                                   | sweep/docking_bench.py          |
| 74  | pdb_entry |       | 5D3L                                   | sweep/docking_bench.py          |
| 75  | pdb_entry |       | 5EWK                                   | sweep/docking_bench.py          |
| 76  | pdb_entry |       | 5EWY                                   | sweep/docking_bench.py          |
| 77  | pdb_entry |       | 5F60                                   | sweep/docking_bench.py          |
| 78  | pdb_entry |       | 5F62                                   | sweep/docking_bench.py          |
| 79  | pdb_entry |       | 5G5V                                   | sweep/docking_bench.py          |
| 80  | pdb_entry |       | 5HTZ                                   | sweep/docking_bench.py          |
| 81  | pdb_entry |       | 5IX0                                   | sweep/docking_bench.py          |
| 82  | pdb_entry |       | 5L8C                                   | sweep/docking_bench.py          |
| 83  | pdb_entry |       | 5MJN                                   | sweep/docking_bench.py          |
| 84  | pdb_entry |       | 5MPN                                   | sweep/docking_bench.py          |
| 85  | pdb_entry |       | 5N24                                   | sweep/docking_bench.py          |
| 86  | pdb_entry |       | 5O87                                   | sweep/docking_bench.py          |
| 87  | pdb_entry |       | 5SZ4                                   | sweep/docking_bench.py          |
| 88  | pdb_entry |       | 5SZ5                                   | sweep/docking_bench.py          |
| 89  | pdb_entry |       | 5Y8Y                                   | sweep/docking_bench.py          |
| 90  | pdb_entry |       | 6EQ5                                   | sweep/docking_bench.py          |
| 91  | pdb_entry |       | 6G38                                   | sweep/docking_bench.py          |
| 92  | pdb_entry |       | 6G6T                                   | sweep/docking_bench.py          |
| 93  | pdb_entry |       | 6HQY                                   | sweep/docking_bench.py          |
| 94  | pdb_entry |       | 6HRQ                                   | sweep/docking_bench.py          |
| 95  | pdb_entry |       | 6OE0                                   | sweep/docking_bench.py          |
| 96  | pdb_entry |       | 6RFN                                   | sweep/docking_bench.py          |
| 97  | pdb_entry |       | 6U6W                                   | sweep/docking_bench.py          |
| 98  | species   |       | 10090-ASTROCYTE-integrated.txt         | sweep/codon_cai_multispecies.py |
| 99  | species   |       | GCF_000001635.27                       | sweep/codon_cai_multispecies.py |
| 100 | species   |       | 10116-CARDIAC_VENTRICLE-integrated.txt | sweep/codon_cai_multispecies.py |
| 101 | species   |       | GCF_036323735.1                        | sweep/codon_cai_multispecies.py |
| 102 | species   |       | 121845-WHOLE_ORGANISM-integrated.txt   | sweep/codon_cai_multispecies.py |
| 103 | species   |       | GCF_000475195.1                        | sweep/codon_cai_multispecies.py |
| 104 | species   |       | 1295-WHOLE_ORGANISM-integrated.txt     | sweep/codon_cai_multispecies.py |
| 105 | species   |       | GCF_025558325.1                        | sweep/codon_cai_multispecies.py |
| 106 | species   |       | 1314-WHOLE_ORGANISM-integrated.txt     | sweep/codon_cai_multispecies.py |
| 107 | species   |       | GCF_900475035.1                        | sweep/codon_cai_multispecies.py |
| 108 | species   |       | 185431-WHOLE_ORGANISM-integrated.txt   | sweep/codon_cai_multispecies.py |
| 109 | species   |       | GCF_000002445.2                        | sweep/codon_cai_multispecies.py |
| 110 | species   |       | 192222-WHOLE_ORGANISM-integrated.txt   | sweep/codon_cai_multispecies.py |
| 111 | species   |       | GCF_000009085.1                        | sweep/codon_cai_multispecies.py |
| 112 | species   |       | 196627-WHOLE_ORGANISM-integrated.txt   | sweep/codon_cai_multispecies.py |
| 113 | species   |       | GCF_000011325.1                        | sweep/codon_cai_multispecies.py |
| 114 | species   |       | 198214-WHOLE_ORGANISM-integrated.txt   | sweep/codon_cai_multispecies.py |
| 115 | species   |       | GCF_000006925.2                        | sweep/codon_cai_multispecies.py |
| 116 | species   |       | 208964-WHOLE_ORGANISM-integrated.txt   | sweep/codon_cai_multispecies.py |
| 117 | species   |       | GCF_000006765.1                        | sweep/codon_cai_multispecies.py |

| #   | Kind    | Study | Accession                                    | Used in                         |
|-----|---------|-------|----------------------------------------------|---------------------------------|
| 118 | species |       | 224308-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 119 | species |       | GCF_000009045.1                              | sweep/codon_cai_multispecies.py |
| 120 | species |       | 224911-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 121 | species |       | GCF_001642675.1                              | sweep/codon_cai_multispecies.py |
| 122 | species |       | 243159-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 123 | species |       | GCF_000021485.1                              | sweep/codon_cai_multispecies.py |
| 124 | species |       | 243230-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 125 | species |       | GCF_020546685.1                              | sweep/codon_cai_multispecies.py |
| 126 | species |       | 246200-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 127 | species |       | GCF_000011965.2                              | sweep/codon_cai_multispecies.py |
| 128 | species |       | 2903-<br>WHOLE_ORGANISM-<br>integrated.txt   | sweep/codon_cai_multispecies.py |
| 129 | species |       | GCF_000372725.1                              | sweep/codon_cai_multispecies.py |
| 130 | species |       | 343509-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 131 | species |       | GCF_000010085.1                              | sweep/codon_cai_multispecies.py |
| 132 | species |       | 36329-<br>WHOLE_ORGANISM-<br>integrated.txt  | sweep/codon_cai_multispecies.py |
| 133 | species |       | GCF_000002765.6                              | sweep/codon_cai_multispecies.py |
| 134 | species |       | 403677-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 135 | species |       | GCF_000142945.1                              | sweep/codon_cai_multispecies.py |
| 136 | species |       | 4081-PERICARP-<br>integrated.txt             | sweep/codon_cai_multispecies.py |
| 137 | species |       | GCF_036512215.1                              | sweep/codon_cai_multispecies.py |
| 138 | species |       | 4097-LEAF-integrated.txt                     | sweep/codon_cai_multispecies.py |
| 139 | species |       | GCF_000715075.1                              | sweep/codon_cai_multispecies.py |
| 140 | species |       | 418985-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/codon_cai_multispecies.py |
| 141 | species |       | GCA_002081605.1                              | sweep/codon_cai_multispecies.py |
| 142 | species |       | 44689-<br>WHOLE_ORGANISM-<br>integrated.txt  | sweep/codon_cai_multispecies.py |
| 143 | species |       | GCF_000004695.1                              | sweep/codon_cai_multispecies.py |
| 144 | species |       | 4932-<br>WHOLE_ORGANISM-<br>integrated.txt   | sweep/codon_cai_multispecies.py |
| 145 | species |       | GCF_000146045.2                              | sweep/codon_cai_multispecies.py |
| 146 | species |       | 54291-<br>WHOLE_ORGANISM-<br>integrated.txt  | sweep/codon_cai_multispecies.py |
| 147 | species |       | GCF_030505655.1                              | sweep/codon_cai_multispecies.py |
| 148 | species |       | 6239-<br>WHOLE_ORGANISM-<br>integrated.txt   | sweep/codon_cai_multispecies.py |
| 149 | species |       | GCF_000002985.6                              | sweep/codon_cai_multispecies.py |
| 150 | species |       | 7227-SPERM-integrated.txt                    | sweep/codon_cai_multispecies.py |
| 151 | species |       | GCF_000001215.4                              | sweep/codon_cai_multispecies.py |
| 152 | species |       | 7460-ABDOMEN-<br>integrated.txt              | sweep/codon_cai_multispecies.py |
| 153 | species |       | GCF_003254395.2                              | sweep/codon_cai_multispecies.py |
| 154 | species |       | 8030-LIVER-integrated.txt                    | sweep/codon_cai_multispecies.py |

| #   | Kind              | Study                | Accession                                               | Used in                                        |
|-----|-------------------|----------------------|---------------------------------------------------------|------------------------------------------------|
| 155 | species           |                      | GCF.965601325.2                                         | sweep/codon_cai_multispecies.py                |
| 156 | species           |                      | 83332-<br>WHOLE_ORGANISM-<br>integrated.txt             | sweep/codon_cai_multispecies.py                |
| 157 | species           |                      | GCF.000195955.2                                         | sweep/codon_cai_multispecies.py                |
| 158 | species           |                      | 9031-HEART-integrated.txt                               | sweep/codon_cai_multispecies.py                |
| 159 | species           |                      | GCF.016699485.2                                         | sweep/codon_cai_multispecies.py                |
| 160 | species           |                      | 93061-<br>WHOLE_ORGANISM-<br>integrated.txt             | sweep/codon_cai_multispecies.py                |
| 161 | species           |                      | GCF.000013425.1                                         | sweep/codon_cai_multispecies.py                |
| 162 | species           |                      | 9544-BILE_DUCT-<br>integrated.txt                       | sweep/codon_cai_multispecies.py                |
| 163 | species           |                      | GCF.049350105.2                                         | sweep/codon_cai_multispecies.py                |
| 164 | species           |                      | 9606-ADRENAL_GLAND-<br>integrated.txt                   | sweep/codon_cai_multispecies.py                |
| 165 | species           |                      | GCF.000001405.40                                        | sweep/codon_cai_multispecies.py                |
| 166 | species           |                      | 9615-PLASMA-<br>integrated.txt                          | sweep/codon_cai_multispecies.py                |
| 167 | species           |                      | GCF.011100685.1                                         | sweep/codon_cai_multispecies.py                |
| 168 | species           |                      | 9823-LIVER-integrated.txt                               | sweep/codon_cai_multispecies.py                |
| 169 | species           |                      | GCF.054392235.1                                         | sweep/codon_cai_multispecies.py                |
| 170 | species           |                      | 9913-<br>LENS_OF_CAMERA_TYPE_EYE-<br>integrated.txt     | sweep/codon_cai_multispecies.py                |
| 171 | species           |                      | GCF.002263795.3                                         | sweep/codon_cai_multispecies.py                |
| 172 | species           |                      | 99287-<br>WHOLE_ORGANISM-<br>integrated.txt             | sweep/codon_cai_multispecies.py                |
| 173 | species           |                      | GCF.000006945.2                                         | sweep/codon_cai_multispecies.py                |
| 174 | species           |                      | 9986-<br>PERIPHERAL_REGION_OF_RETINA-<br>integrated.txt | sweep/codon_cai_multispecies.py                |
| 175 | species           |                      | GCF.964237555.1                                         | sweep/codon_cai_multispecies.py                |
| 176 | gtrnadb           |                      | Esch.coli_K_12_MG1655                                   | sweep/codon_tai_enc.py                         |
| 177 | gtrnadb           |                      | Baci.subt_subtilis_168                                  | sweep/codon_tai_enc.py                         |
| 178 | gtrnadb           |                      | Scere3                                                  | sweep/codon_tai_enc.py                         |
| 179 | refseq_transcript | 25                   | NM.000546.6                                             | sweep/primer_vs_primer3.py                     |
| 180 | refseq_transcript | 25                   | NM.007294.4                                             | sweep/primer_vs_primer3.py                     |
| 181 | refseq_transcript | 25                   | NM.000059.4                                             | sweep/primer_vs_primer3.py                     |
| 182 | refseq_transcript | 25                   | NM.004333.6                                             | sweep/primer_vs_primer3.py                     |
| 183 | refseq_transcript | 25                   | NM.005228.5                                             | sweep/primer_vs_primer3.py                     |
| 184 | study             | 26                   | clinvar.vcf.gz                                          | sweep/vcf_vs_pysam.py                          |
| 185 | study             | 27                   | GSE52778                                                | sweep/de_vs_pydeseq2.py                        |
| 186 | rfam_family       | RF00004              | RF00004                                                 | sweep/stockholm_vs_biopython.py                |
| 187 | rfam_family       | RF00015              | RF00015                                                 | sweep/stockholm_vs_biopython.py                |
| 188 | rfam_family       | RF00017              | RF00017                                                 | sweep/stockholm_vs_biopython.py                |
| 189 | rfam_family       | RF00023              | RF00023                                                 | sweep/stockholm_vs_biopython.py                |
| 190 | rfam_family       | RF00050              | RF00050                                                 | sweep/stockholm_vs_biopython.py                |
| 191 | rfam_family       | RF00059              | RF00059                                                 | sweep/stockholm_vs_biopython.py                |
| 192 | rfam_family       | RF00162              | RF00162                                                 | sweep/stockholm_vs_biopython.py                |
| 193 | rfam_family       | RF00167              | RF00167                                                 | sweep/stockholm_vs_biopython.py                |
| 194 | rfam_family       | RF00168              | RF00168                                                 | sweep/stockholm_vs_biopython.py                |
| 195 | rfam_family       | RF00174              | RF00174                                                 | sweep/stockholm_vs_biopython.py                |
| 196 | rfam_family       | RF00234              | RF00234                                                 | sweep/stockholm_vs_biopython.py                |
| 197 | rfam_family       | RF00504              | RF00504                                                 | sweep/stockholm_vs_biopython.py                |
| 198 | refseq_genome     | J01749.1             | J01749.1                                                | sweep/proteinprops_restriction_vs_biopython.py |
| 199 | refseq_genome     | NC_001416.1          | NC_001416.1                                             | sweep/proteinprops_restriction_vs_biopython.py |
| 200 | opentree_node     | ott1062253           | ott1062253                                              | sweep/newick_vs_biophylo.py                    |
| 201 | opentree_node     | ott392223            | ott392223                                               | sweep/newick_vs_biophylo.py                    |
| 202 | opentree_node     | ott44565             | ott44565                                                | sweep/newick_vs_biophylo.py                    |
| 203 | opentree_node     | ott691846            | ott691846                                               | sweep/newick_vs_biophylo.py                    |
| 204 | opentree_node     | ott81461             | ott81461                                                | sweep/newick_vs_biophylo.py                    |
| 205 | opentree_node     | ott913935            | ott913935                                               | sweep/newick_vs_biophylo.py                    |
| 206 | study             | NA12878_chr20_lowcov | NA12878.chrom20.ILLUMINA                                | sweep/fit_low_pysam.py                         |
| 207 | study             | SRR622461            | SRR622461                                               | sweep/fastq_vs_biopython.py                    |
| 208 | jaspar_matrix     | MA0003.4             | MA0003.4                                                | sweep/motif_vs_biopython.py                    |

| #   | Kind           | Study          | Accession                | Used in                       |
|-----|----------------|----------------|--------------------------|-------------------------------|
| 209 | jaspar_matrix  | MA0079.3       | MA0079.3                 | sweep/motif_vs.biopython.py   |
| 210 | jaspar_matrix  | MA0080.4       | MA0080.4                 | sweep/motif_vs.biopython.py   |
| 211 | jaspar_matrix  | MA0099.3       | MA0099.3                 | sweep/motif_vs.biopython.py   |
| 212 | jaspar_matrix  | MA0105.4       | MA0105.4                 | sweep/motif_vs.biopython.py   |
| 213 | jaspar_matrix  | MA0106.3       | MA0106.3                 | sweep/motif_vs.biopython.py   |
| 214 | jaspar_matrix  | MA0108.2       | MA0108.2                 | sweep/motif_vs.biopython.py   |
| 215 | jaspar_matrix  | MA0112.3       | MA0112.3                 | sweep/motif_vs.biopython.py   |
| 216 | jaspar_matrix  | MA0113.3       | MA0113.3                 | sweep/motif_vs.biopython.py   |
| 217 | jaspar_matrix  | MA0139.1       | MA0139.1                 | sweep/motif_vs.biopython.py   |
| 218 | nucore_record  | NC_000913.3    | NC_000913.3              | sweep/genbank_vs.biopython.py |
| 219 | nucore_record  | NC_001422.1    | NC_001422.1              | sweep/genbank_vs.biopython.py |
| 220 | nucore_record  | NC_001477.1    | NC_001477.1              | sweep/genbank_vs.biopython.py |
| 221 | nucore_record  | NC_001498.1    | NC_001498.1              | sweep/genbank_vs.biopython.py |
| 222 | nucore_record  | NC_001802.1    | NC_001802.1              | sweep/genbank_vs.biopython.py |
| 223 | nucore_record  | NC_001806.2    | NC_001806.2              | sweep/genbank_vs.biopython.py |
| 224 | nucore_record  | NC_002549.1    | NC_002549.1              | sweep/genbank_vs.biopython.py |
| 225 | nucore_record  | NC_003977.2    | NC_003977.2              | sweep/genbank_vs.biopython.py |
| 226 | nucore_record  | NC_012920.1    | NC_012920.1              | sweep/genbank_vs.biopython.py |
| 227 | nucore_record  | NC_045512.2    | NC_045512.2              | sweep/genbank_vs.biopython.py |
| 228 | nucore_record  | NG_005905.2    | NG_005905.2              | sweep/genbank_vs.biopython.py |
| 229 | nucore_record  | NM_000038.6    | NM_000038.6              | sweep/genbank_vs.biopython.py |
| 230 | nucore_record  | NM_000249.4    | NM_000249.4              | sweep/genbank_vs.biopython.py |
| 231 | nucore_record  | NM_000251.3    | NM_000251.3              | sweep/genbank_vs.biopython.py |
| 232 | nucore_record  | NM_000314.8    | NM_000314.8              | sweep/genbank_vs.biopython.py |
| 233 | nucore_record  | NM_001126112.3 | NM_001126112.3           | sweep/genbank_vs.biopython.py |
| 234 | nucore_record  | NM_001904.4    | NM_001904.4              | sweep/genbank_vs.biopython.py |
| 235 | nucore_record  | NM_002524.5    | NM_002524.5              | sweep/genbank_vs.biopython.py |
| 236 | nucore_record  | NM_004985.5    | NM_004985.5              | sweep/genbank_vs.biopython.py |
| 237 | study          | 29             | hg38.cpgIslandExt.chr20  | sweep/bed_vs.bioframe.py      |
| 238 | study          | 30             | hg38.rmsk.chr20:0-500000 | sweep/bed_vs.bioframe.py      |
| 239 | nucore_record  | MG772933.1     | MG772933.1               | sweep/kmer_vs.sourmash.py     |
| 240 | nucore_record  | MN996532.2     | MN996532.2               | sweep/kmer_vs.sourmash.py     |
| 241 | nucore_record  | MT040333.1     | MT040333.1               | sweep/kmer_vs.sourmash.py     |
| 242 | nucore_record  | NC_004718.3    | NC_004718.3              | sweep/kmer_vs.sourmash.py     |
| 243 | nucore_record  | NC_019843.3    | NC_019843.3              | sweep/kmer_vs.sourmash.py     |
| 244 | db SNP_variant | rs1042713      | rs1042713                | sweep/gstats_vs.scipy.py      |
| 245 | db SNP_variant | rs10427255     | rs10427255               | sweep/gstats_vs.scipy.py      |
| 246 | db SNP_variant | rs1229984      | rs1229984                | sweep/gstats_vs.scipy.py      |
| 247 | db SNP_variant | rs12913832     | rs12913832               | sweep/gstats_vs.scipy.py      |
| 248 | db SNP_variant | rs1426654      | rs1426654                | sweep/gstats_vs.scipy.py      |
| 249 | db SNP_variant | rs16891982     | rs16891982               | sweep/gstats_vs.scipy.py      |
| 250 | db SNP_variant | rs17822931     | rs17822931               | sweep/gstats_vs.scipy.py      |
| 251 | db SNP_variant | rs1799945      | rs1799945                | sweep/gstats_vs.scipy.py      |
| 252 | db SNP_variant | rs1800562      | rs1800562                | sweep/gstats_vs.scipy.py      |
| 253 | db SNP_variant | rs1801133      | rs1801133                | sweep/gstats_vs.scipy.py      |
| 254 | db SNP_variant | rs1815739      | rs1815739                | sweep/gstats_vs.scipy.py      |
| 255 | db SNP_variant | rs2231142      | rs2231142                | sweep/gstats_vs.scipy.py      |
| 256 | db SNP_variant | rs334          | rs334                    | sweep/gstats_vs.scipy.py      |
| 257 | db SNP_variant | rs3827760      | rs3827760                | sweep/gstats_vs.scipy.py      |
| 258 | db SNP_variant | rs429358       | rs429358                 | sweep/gstats_vs.scipy.py      |
| 259 | db SNP_variant | rs4680         | rs4680                   | sweep/gstats_vs.scipy.py      |
| 260 | db SNP_variant | rs4988235      | rs4988235                | sweep/gstats_vs.scipy.py      |
| 261 | db SNP_variant | rs53576        | rs53576                  | sweep/gstats_vs.scipy.py      |
| 262 | db SNP_variant | rs6025         | rs6025                   | sweep/gstats_vs.scipy.py      |
| 263 | db SNP_variant | rs671          | rs671                    | sweep/gstats_vs.scipy.py      |
| 264 | db SNP_variant | rs7412         | rs7412                   | sweep/gstats_vs.scipy.py      |
| 265 | db SNP_variant | rs762551       | rs762551                 | sweep/gstats_vs.scipy.py      |
| 266 | hgnc_record    | HGNC:ABCC11    | HGNC:14639               | sweep/uniprot_vs.hgnc.py      |
| 267 | hgnc_record    | HGNC:ABL1      | HGNC:76                  | sweep/uniprot_vs.hgnc.py      |
| 268 | hgnc_record    | HGNC:ACTN3     | HGNC:165                 | sweep/uniprot_vs.hgnc.py      |
| 269 | hgnc_record    | HGNC:ADH1B     | HGNC:250                 | sweep/uniprot_vs.hgnc.py      |
| 270 | hgnc_record    | HGNC:ALDH2     | HGNC:404                 | sweep/uniprot_vs.hgnc.py      |
| 271 | hgnc_record    | HGNC:APC       | HGNC:583                 | sweep/uniprot_vs.hgnc.py      |
| 272 | hgnc_record    | HGNC:APOE      | HGNC:613                 | sweep/uniprot_vs.hgnc.py      |
| 273 | hgnc_record    | HGNC:ASXL1     | HGNC:18318               | sweep/uniprot_vs.hgnc.py      |
| 274 | hgnc_record    | HGNC:ATM       | HGNC:795                 | sweep/uniprot_vs.hgnc.py      |

| #   | Kind        | Study        | Accession                              | Used in                                        |
|-----|-------------|--------------|----------------------------------------|------------------------------------------------|
| 275 | hgnc_record | HGNC:BRAF    | HGNC:1097                              | sweep/uniprot_vs_hgnc.py                       |
| 276 | hgnc_record | HGNC:BRCA1   | HGNC:1100                              | sweep/uniprot_vs_hgnc.py                       |
| 277 | hgnc_record | HGNC:BRCA2   | HGNC:1101                              | sweep/uniprot_vs_hgnc.py                       |
| 278 | hgnc_record | HGNC:CDKN2A  | HGNC:1787                              | sweep/uniprot_vs_hgnc.py                       |
| 279 | hgnc_record | HGNC:CFTR    | HGNC:1884                              | sweep/uniprot_vs_hgnc.py                       |
| 280 | hgnc_record | HGNC:CHEK2   | HGNC:16627                             | sweep/uniprot_vs_hgnc.py                       |
| 281 | hgnc_record | HGNC:COMT    | HGNC:2228                              | sweep/uniprot_vs_hgnc.py                       |
| 282 | hgnc_record | HGNC:DMD     | HGNC:2928                              | sweep/uniprot_vs_hgnc.py                       |
| 283 | hgnc_record | HGNC:DNMT3A  | HGNC:2978                              | sweep/uniprot_vs_hgnc.py                       |
| 284 | hgnc_record | HGNC:EDAR    | HGNC:2895                              | sweep/uniprot_vs_hgnc.py                       |
| 285 | hgnc_record | HGNC:EGFR    | HGNC:3236                              | sweep/uniprot_vs_hgnc.py                       |
| 286 | hgnc_record | HGNC:ERBB2   | HGNC:3430                              | sweep/uniprot_vs_hgnc.py                       |
| 287 | hgnc_record | HGNC:F5      | HGNC:3542                              | sweep/uniprot_vs_hgnc.py                       |
| 288 | hgnc_record | HGNC:FLT3    | HGNC:3765                              | sweep/uniprot_vs_hgnc.py                       |
| 289 | hgnc_record | HGNC:FMR1    | HGNC:3775                              | sweep/uniprot_vs_hgnc.py                       |
| 290 | hgnc_record | HGNC:GBA1    | HGNC:4177                              | sweep/uniprot_vs_hgnc.py                       |
| 291 | hgnc_record | HGNC:HBB     | HGNC:4827                              | sweep/uniprot_vs_hgnc.py                       |
| 292 | hgnc_record | HGNC:HERC2   | HGNC:4868                              | sweep/uniprot_vs_hgnc.py                       |
| 293 | hgnc_record | HGNC:HFE     | HGNC:4886                              | sweep/uniprot_vs_hgnc.py                       |
| 294 | hgnc_record | HGNC:HTT     | HGNC:4851                              | sweep/uniprot_vs_hgnc.py                       |
| 295 | hgnc_record | HGNC:IDH1    | HGNC:5382                              | sweep/uniprot_vs_hgnc.py                       |
| 296 | hgnc_record | HGNC:IDH2    | HGNC:5383                              | sweep/uniprot_vs_hgnc.py                       |
| 297 | hgnc_record | HGNC:JAK2    | HGNC:6192                              | sweep/uniprot_vs_hgnc.py                       |
| 298 | hgnc_record | HGNC:KCNQ1   | HGNC:6294                              | sweep/uniprot_vs_hgnc.py                       |
| 299 | hgnc_record | HGNC:KRAS    | HGNC:6407                              | sweep/uniprot_vs_hgnc.py                       |
| 300 | hgnc_record | HGNC:LCT     | HGNC:6530                              | sweep/uniprot_vs_hgnc.py                       |
| 301 | hgnc_record | HGNC:LMNA    | HGNC:6636                              | sweep/uniprot_vs_hgnc.py                       |
| 302 | hgnc_record | HGNC:MLH1    | HGNC:7127                              | sweep/uniprot_vs_hgnc.py                       |
| 303 | hgnc_record | HGNC:MSH2    | HGNC:7325                              | sweep/uniprot_vs_hgnc.py                       |
| 304 | hgnc_record | HGNC:MTHFR   | HGNC:7436                              | sweep/uniprot_vs_hgnc.py                       |
| 305 | hgnc_record | HGNC:MYC     | HGNC:7553                              | sweep/uniprot_vs_hgnc.py                       |
| 306 | hgnc_record | HGNC:MYH7    | HGNC:7577                              | sweep/uniprot_vs_hgnc.py                       |
| 307 | hgnc_record | HGNC:NF1     | HGNC:7765                              | sweep/uniprot_vs_hgnc.py                       |
| 308 | hgnc_record | HGNC:NF2     | HGNC:7773                              | sweep/uniprot_vs_hgnc.py                       |
| 309 | hgnc_record | HGNC:NPM1    | HGNC:7910                              | sweep/uniprot_vs_hgnc.py                       |
| 310 | hgnc_record | HGNC:NRAS    | HGNC:7989                              | sweep/uniprot_vs_hgnc.py                       |
| 311 | hgnc_record | HGNC:OCA2    | HGNC:8101                              | sweep/uniprot_vs_hgnc.py                       |
| 312 | hgnc_record | HGNC:PALB2   | HGNC:26144                             | sweep/uniprot_vs_hgnc.py                       |
| 313 | hgnc_record | HGNC:PIK3CA  | HGNC:8975                              | sweep/uniprot_vs_hgnc.py                       |
| 314 | hgnc_record | HGNC:PTEN    | HGNC:9588                              | sweep/uniprot_vs_hgnc.py                       |
| 315 | hgnc_record | HGNC:RB1     | HGNC:9884                              | sweep/uniprot_vs_hgnc.py                       |
| 316 | hgnc_record | HGNC:RUNX1   | HGNC:10471                             | sweep/uniprot_vs_hgnc.py                       |
| 317 | hgnc_record | HGNC:SCN5A   | HGNC:10593                             | sweep/uniprot_vs_hgnc.py                       |
| 318 | hgnc_record | HGNC:SLC24A5 | HGNC:20611                             | sweep/uniprot_vs_hgnc.py                       |
| 319 | hgnc_record | HGNC:SMAD4   | HGNC:6770                              | sweep/uniprot_vs_hgnc.py                       |
| 320 | hgnc_record | HGNC:SOD1    | HGNC:11179                             | sweep/uniprot_vs_hgnc.py                       |
| 321 | hgnc_record | HGNC:STK11   | HGNC:11389                             | sweep/uniprot_vs_hgnc.py                       |
| 322 | hgnc_record | HGNC:TET2    | HGNC:25941                             | sweep/uniprot_vs_hgnc.py                       |
| 323 | hgnc_record | HGNC:TP53    | HGNC:11998                             | sweep/uniprot_vs_hgnc.py                       |
| 324 | hgnc_record | HGNC:TTN     | HGNC:12403                             | sweep/uniprot_vs_hgnc.py                       |
| 325 | hgnc_record | HGNC:VHL     | HGNC:12687                             | sweep/uniprot_vs_hgnc.py                       |
| 326 | study       | 31           | UniProt2Reactome.txt (current release) | sweep/reactome_vs_download.py                  |
| 327 | study       | 32           | gnomad_r4<br>gnomad.constraint         | gnomad<br>sweep/gnomad_constraint_threshold.py |

## References

- [1] Doench JG et al. Optimized sgRNA design to maximize activity and minimize off-target effects of CRISPR-Cas9. *Nat Biotechnol* 34:184-191, 2016.
- [2] Concordet JP, Haeussler M. CRISPOR: intuitive guide selection for CRISPR/Cas9 genome editing experiments and screens. *Nucleic Acids Res* 46:W242-W245, 2018.

- [3] Jankauskaitė J et al. SKEMPI 2.0: an updated benchmark of changes in protein-protein binding energy, kinetics and thermodynamics upon mutation. *Bioinformatics* 35:462-469, 2019.
- [4] King ZA et al. BiGG Models: a platform for integrating, standardizing and sharing genome-scale models. *Nucleic Acids Res* 44:D515-522, 2016.
- [5] Sievers F et al. Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega. *Mol Syst Biol* 7:539, 2011.
- [6] Tavtigian SV et al. Modeling the ACMG/AMP variant classification guidelines as a Bayesian classification framework. *Genet Med* 20:1054-1060, 2018.
- [7] Yeo G, Burge CB. Maximum entropy modeling of short sequence motifs with applications to RNA splicing signals. *J Comput Biol* 11:377-394 (2004).
- [8] Sharp PM, Li WH. The codon adaptation index: a measure of directional synonymous codon usage bias. *Nucleic Acids Res* 15:1281-1295 (1987).
- [9] dos Reis M, Savva R, Wernisch L. Solving the riddle of codon usage preferences: a test for translational selection. *Nucleic Acids Res* 32:5036-5044 (2004).
- [10] Eberhardt J, Santos-Martins D, Tillack AF, Forli S. AutoDock Vina 1.2.0. *J Chem Inf Model* 61:3891-3898 (2021).
- [11] Wang M et al. PaxDb, a database of protein abundance averages across all three domains of life. *Mol Cell Proteomics* 11:492-500 (2012).